pebble.

prototype testing

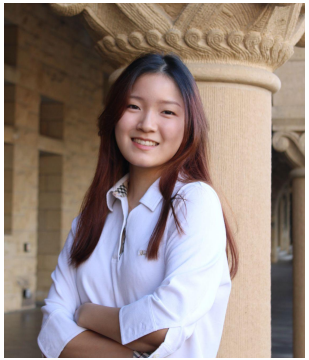
Team 4

Assignment 5
cs 147 autumn 2026

Team Members

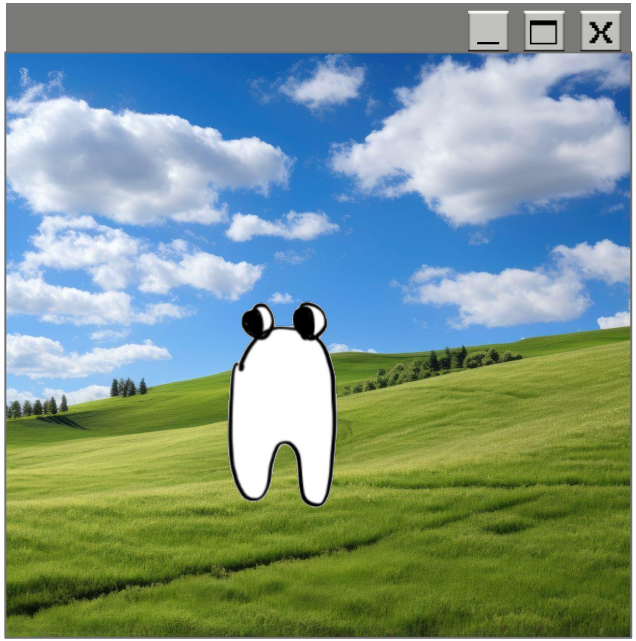
Ryan L.

Brian L.



Cherie C.

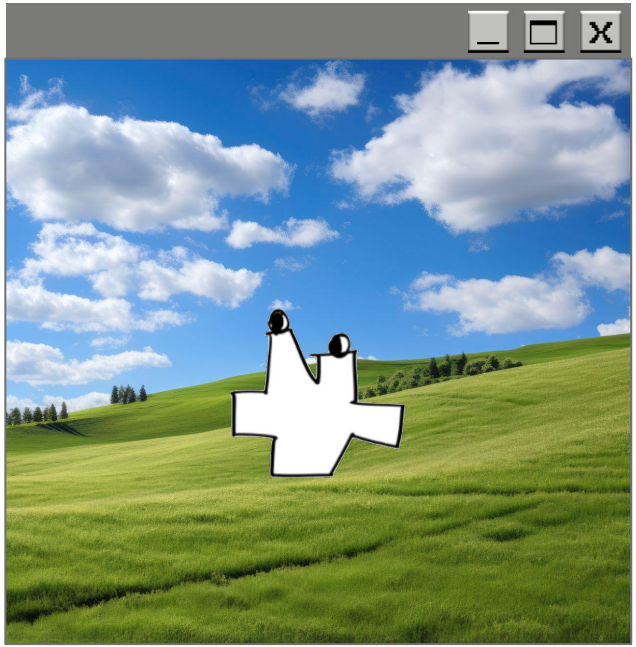
SLIDESGO



pebble.

imperfection rocks

- We want users to slowly accumulate **imperfect** drawings until they create a greater artifact (their gallery) that is sharable with other users
- The value proposition has a double meaning: we call user drawings “pebbles”, and since they are not perfect, slogan is **“imperfection rocks”**



pebble.

imperfection rocks

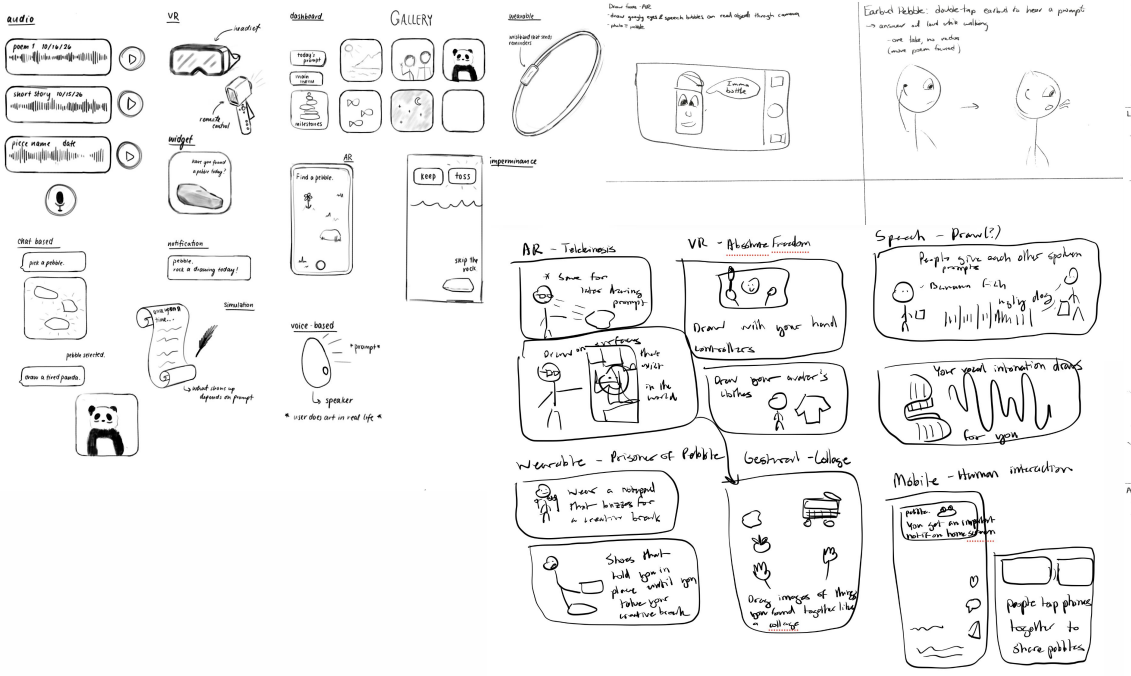
→ Daily timed drawing prompts and platform for **busy** students and professionals with a hunger for creativity but no time

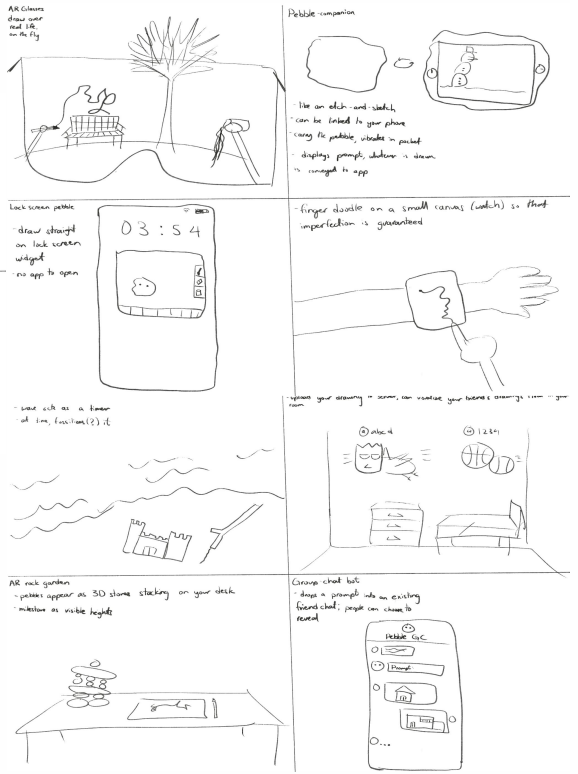
→ See **other users' drawings** for the same prompt after completion, invite friends, and reach milestones

01 Sketches

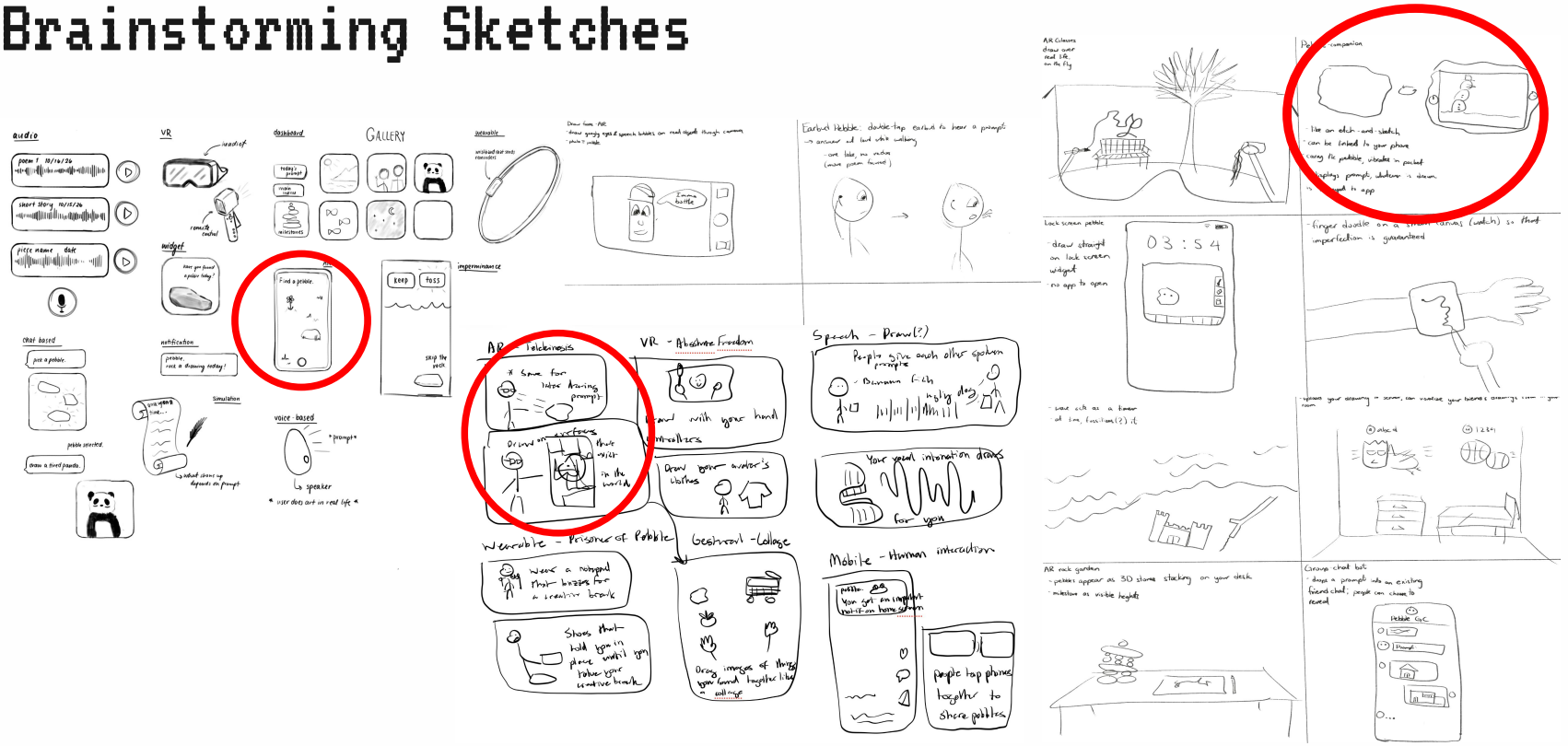
Brainstorming, Concept, Our Fleshed-Out Realizations

Brainstorming Sketches

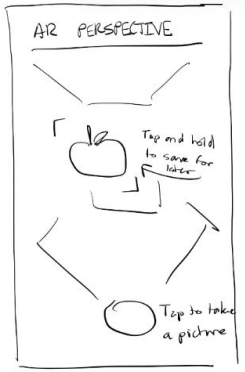




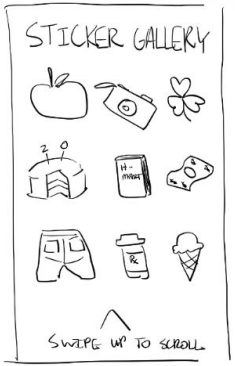
Brainstorming Sketches



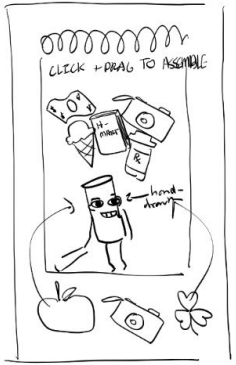
Concept Sketches - AR Realization



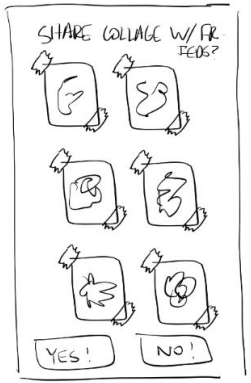
User goes about their day and finds object of interest



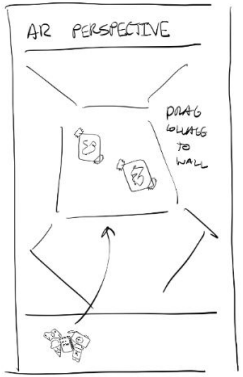
User can look at their previously stored stickers to gain inspiration/reminisce on intentionally important objects.



User gets a notification on their phone that it's time to build their pebble. They can use previously scavenged (?) sticks and draw in relation to them.



User can choose to share their sketch/collage with their friends (social aspect)

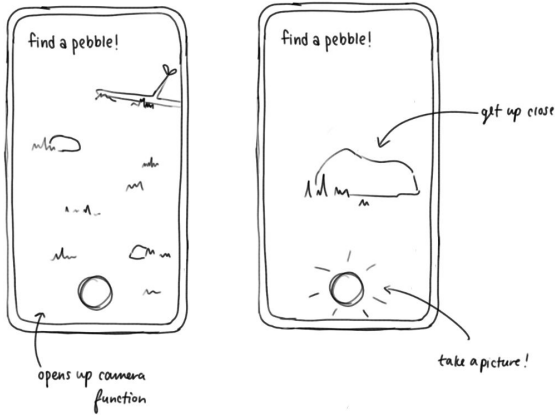


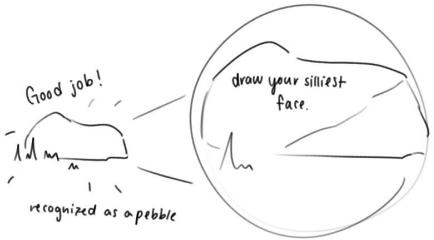
User can place memories on geo-cached room locations

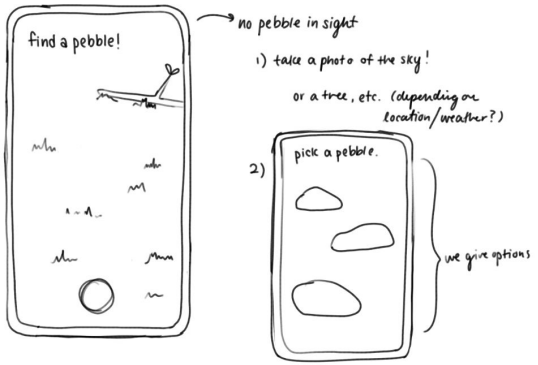
User can flip through all their collages

Concept Sketches - Physical Product Realization

Concept Sketches - Camera Work Realization



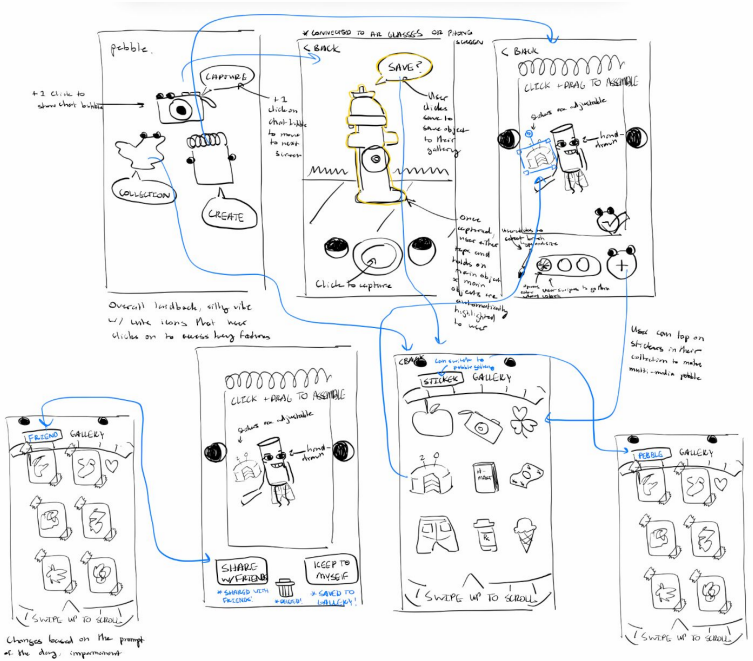




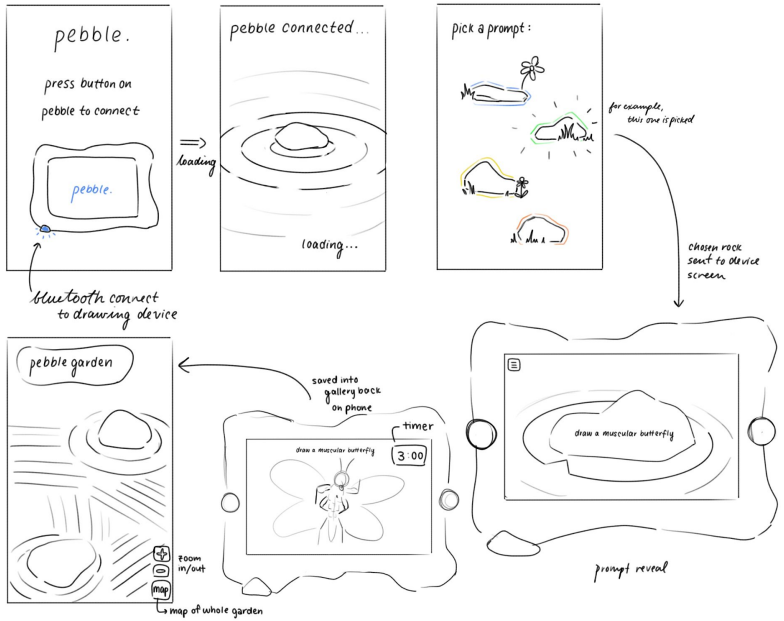
Our Focused Realizations

Our 2 Focused Realizations

01

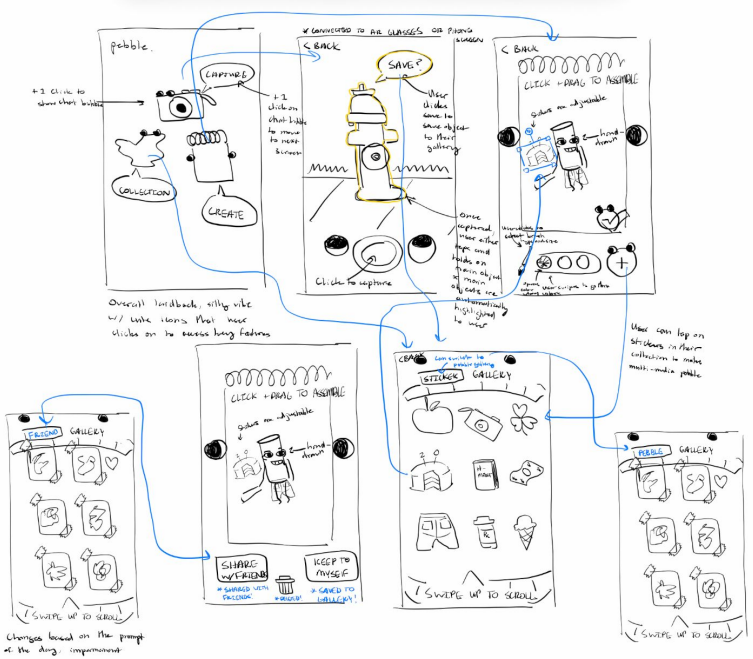


02



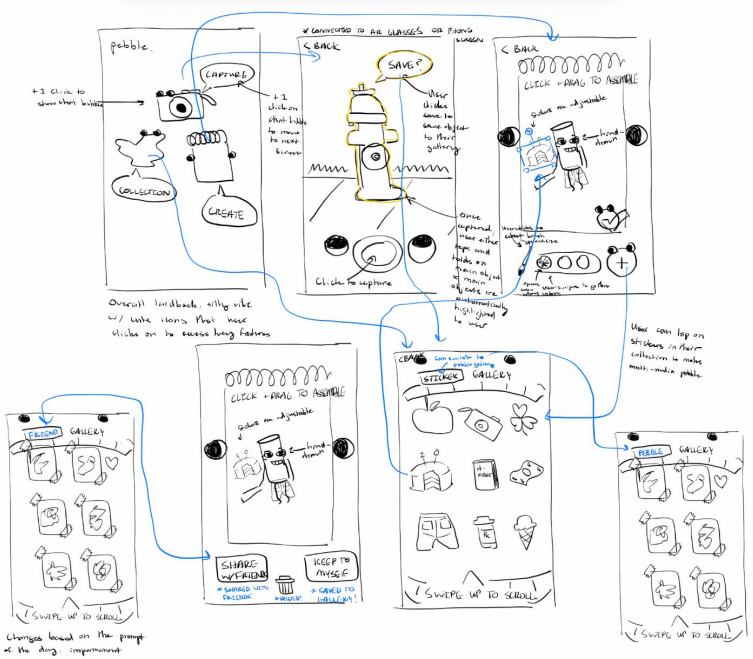
Selected
Interface
and Rationale
OK
Cancel
Apply

AR Sticker Collage: Pros



1. Interactive with the real world
2. Low barrier to entry
3. Reusable stickers
4. Allow easy share options with friends

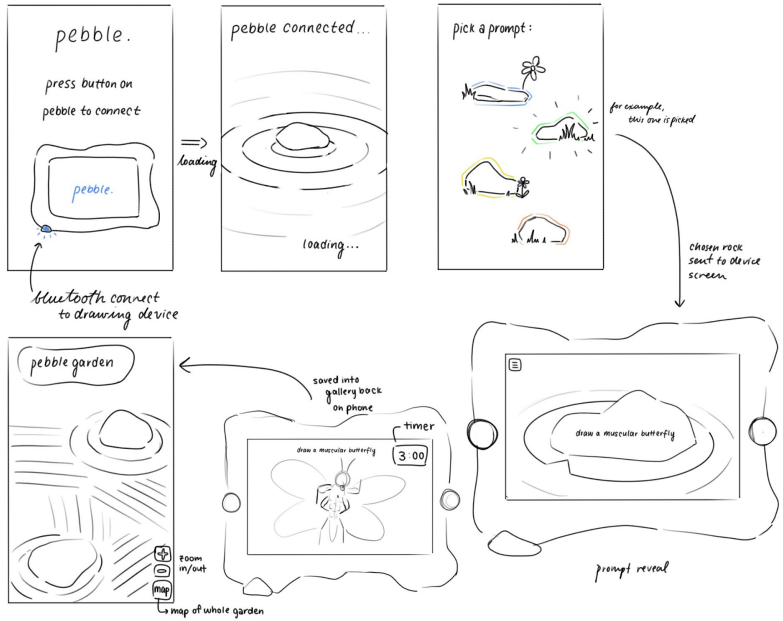
AR Sticker Collage: Cons



1. Harder to implement drawing element
2. Content moderation risk
3. Weaker fit for our needfinding and initial tasks
4. Heavily visual – some accessibility concerns

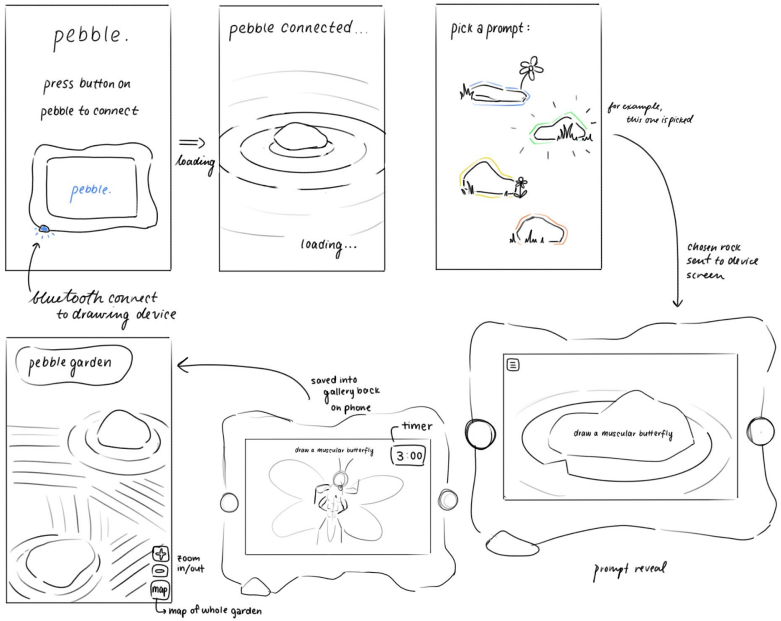
Physical Device: Pros

1. Matches our needfinding
2. Dedicated device -> fewer distractions
3. Timer and random prompts lower the pressure
4. Not mentally demanding



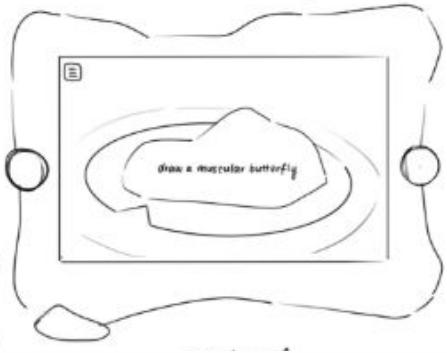
Physical Device: Cons

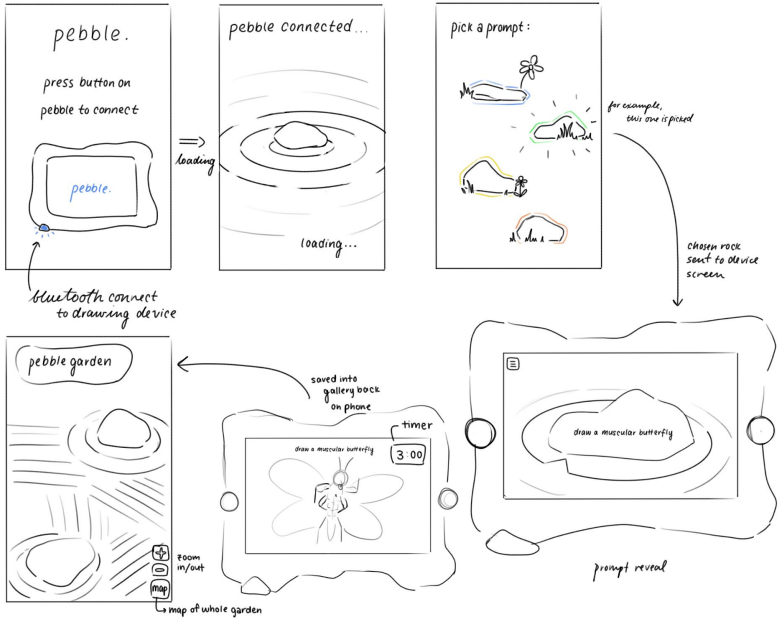
1. Must buy the device to start
2. Sharing requires friends to own one too
3. Easy to leave behind
4. Drawing only



What did we go with?

The physical pebble device.





Why pebble device over AR?

AR app: its cons hit the core loop

Drawing on a phone is awkward → drift photo-to-collage (assembling instead of making)

Hunting for real-world inspiration → effort and social awkwardness (opposite of low-stakes)

Without the AR layer → standard drawing-sharing app that maps poorly to our needfinding and tasks

device: its cons are peripheral

Missing social layer was addable → shared pebble garden and invite flow

Drawing-only scope fits the quick, deliberately silly sketch premise

Randomized prompts and a short timer lower the stakes

trade off we accepted

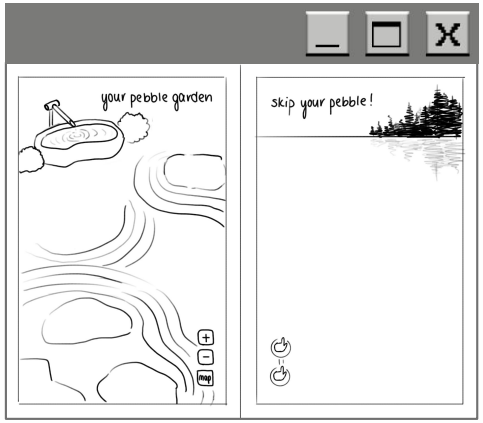
An extra device isn't technically required

Physical separation *is* the refuge → users off the device where their work lives

A second device for a protected break ✓

AR: richer capture and social features at the cost of the break

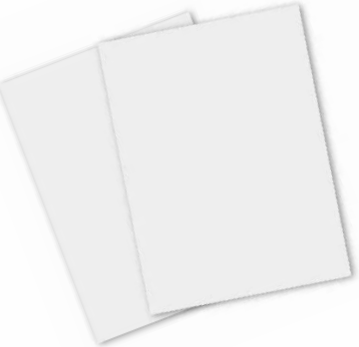
01 Lo-Fi Prototype Construction



Initially designed digitally

(Goodnotes, Procreate, etc.)

- Easy to share with each other
- Easy to edit without wasting paper



Then we printed everything out

- Physicality to prototype
- Easy to interact with

Lo-Fi Prototype

Task Flows

Initial Tasks

Simple:
- Take a quick creative break outside of work
- Pick a prompt and complete a drawing

Moderate:
- Make room for creativity on your own terms
- Reach mile-”stones”

Complex:
- Keep your creative practice alive through a busy stretch
- Invite a friend to the platform

Tasks We Added

Simple:
- Connect a device
- Keep or toss the drawing

Moderate:
- View gallery of previous work
- Share a drawing with friends

Complex:
- Invite a friend to a shared gallery
- Set a personal timer
- Edit personal gallery

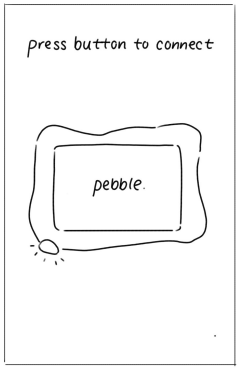
Aggregate Task Flow: Connect Device and Complete a Drawing

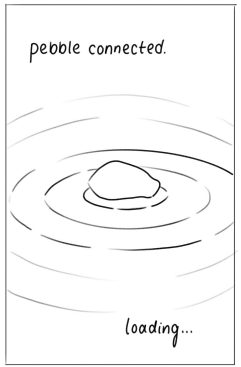
Connect phone to device

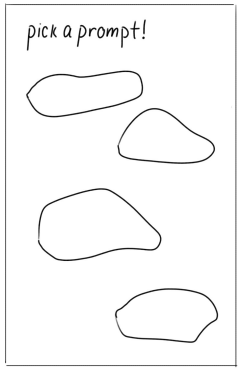
Pick a prompt

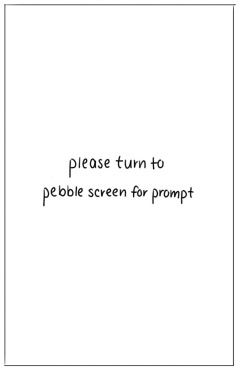
Do the drawing

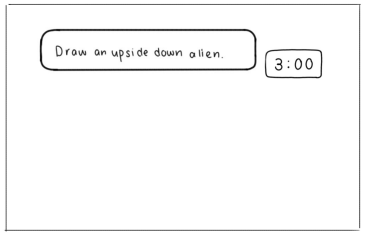
Keep or toss

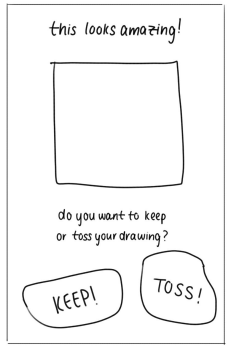




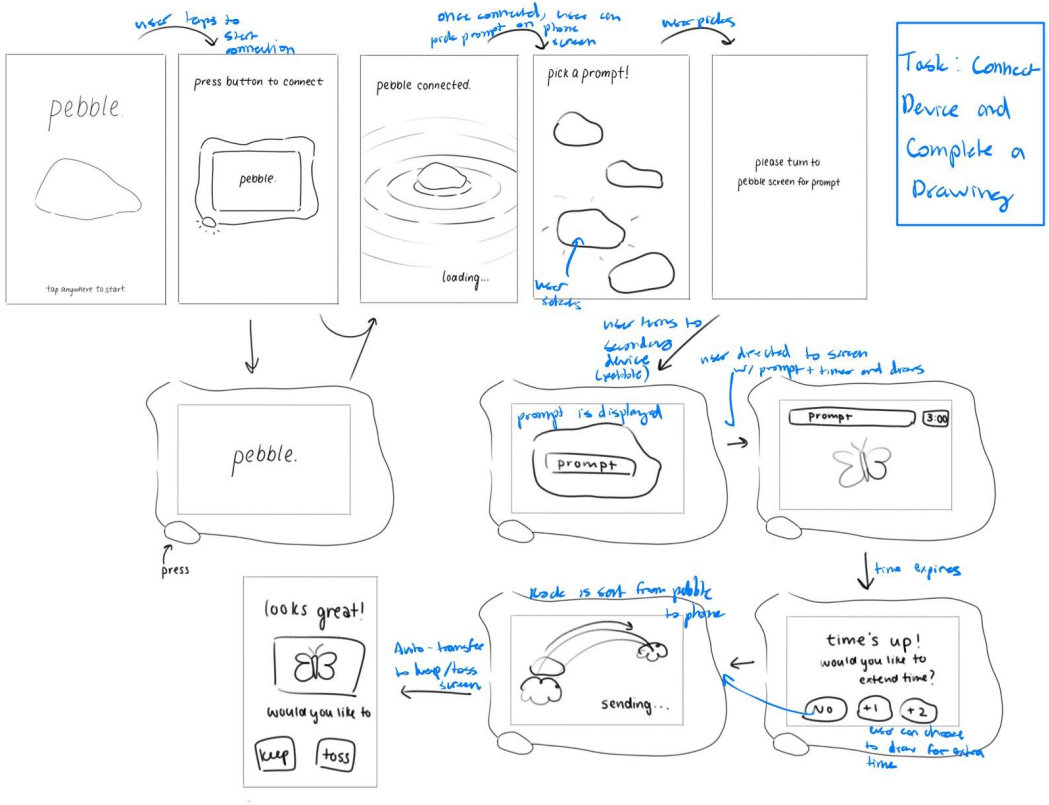


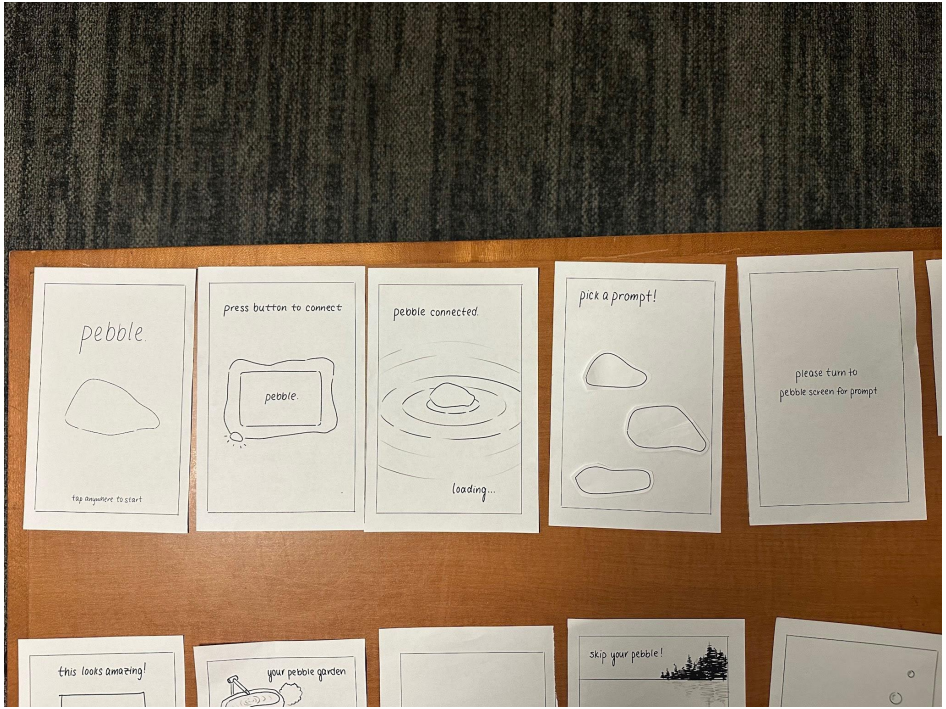






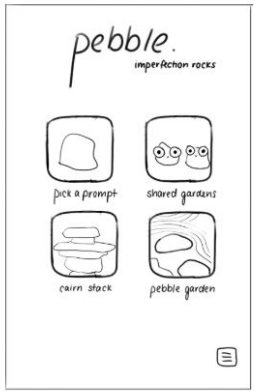
SLIDESGO


user taps to start connection
once connected, user can pick prompt on phone screen
user picks
pebble.
tap anywhere to start
press button to connect
pebble.
pebble connected.
loading...
pick a prompt!
user selects
please turn to pebble screen for prompt
Task: Connect Device and Complete a Drawing
pebble.
press
user turns to recording device (pebble)
prompt is displayed
prompt
user directed to screen w/ prompt + timer and draws
prompt
3:00
time expires
time's up!
would you like to extend time?
No
+1
+2
user can choose to draw for extra time
sketch is sent from pebble to phone
sending...
auto-transfer to keep/toss screen
looks great!
would you like to
keep
toss

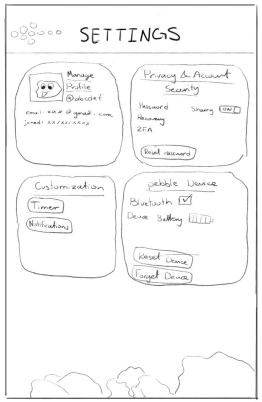

pebble.
tap anywhere to start
press button to connect
pebble.
pebble connected.
loading...
pick a prompt!
please turn to
pebble screen for prompt
this looks amazing!
your pebble garden
skip your pebble!

Aggregate Task Flow: Change the Timer Setting

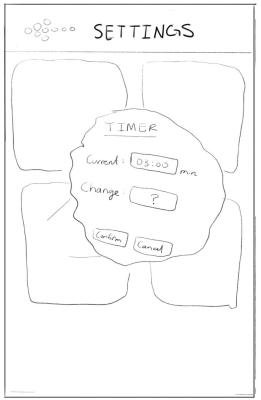
Start at home page



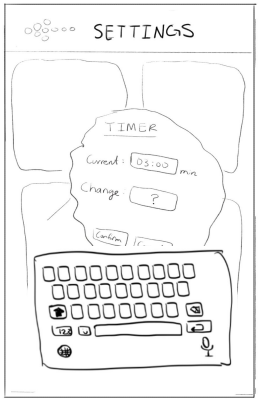
Go to Settings



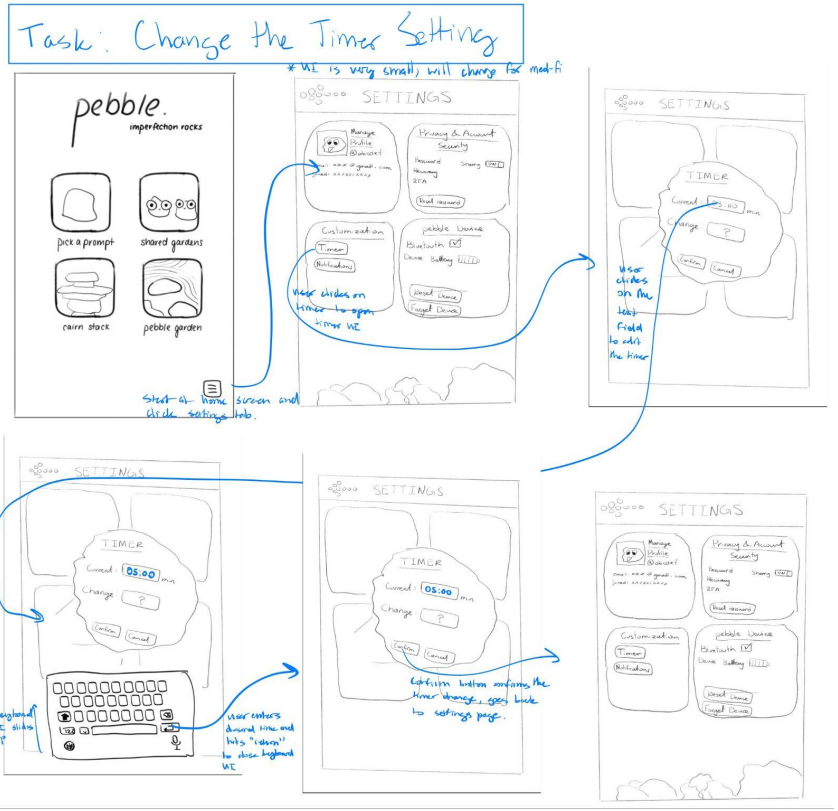
Select Timer Settings



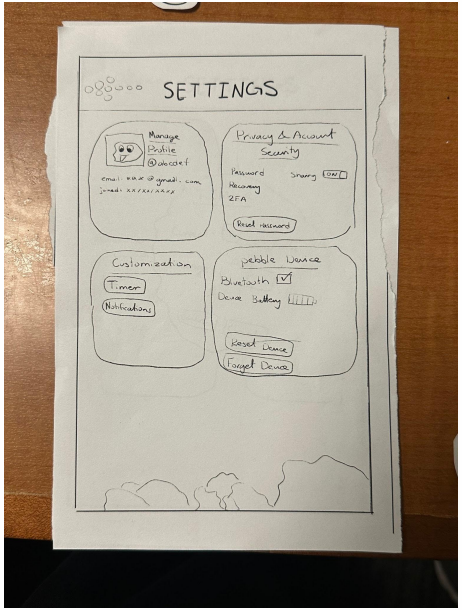
Change the Time

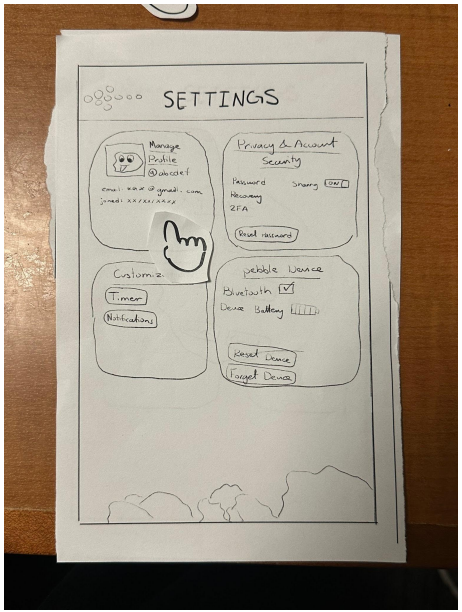


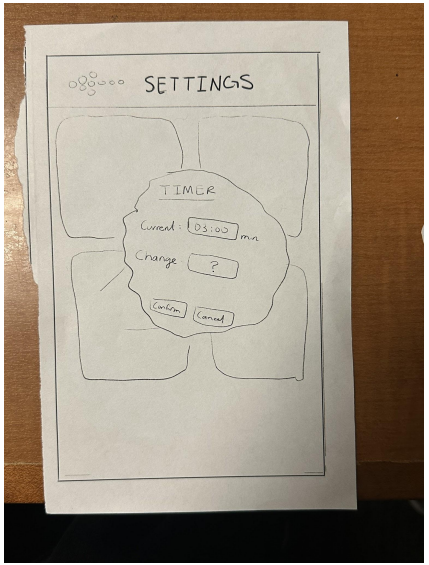
SLIDESGO


Task: Change the Timer Setting
* UI is very small, will change for mod-fi
pebble.
imperfection rocks
pick a prompt
shared gardens
cairn stack
pebble garden
Start at home screen and click Settings tab.
SETTINGS
Manage Profile
Privacy & Account Security
Customization
Timer
Notifications
pebble Device
Bluetooth
Device Battery
Reset Device
Forget Device
Click circles on timer to open timer UI
TIMER
Current: 05:00 min
Change
Confirm
Cancel
User clicks on the text field to edit the timer
Keyboard UI slides up
User enters desired time and hits "change" to close keyboard UI
Confirm button confirms the timer change, goes back to settings page.

Changing the timer

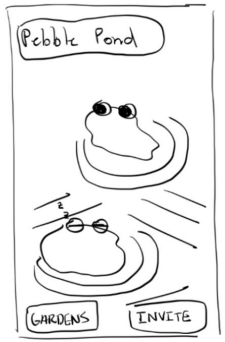






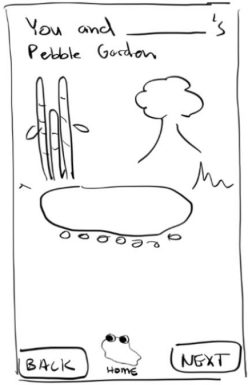
Aggregate Task Flow: Create and Explore a Shared Garden

Invite someone to a shared gallery



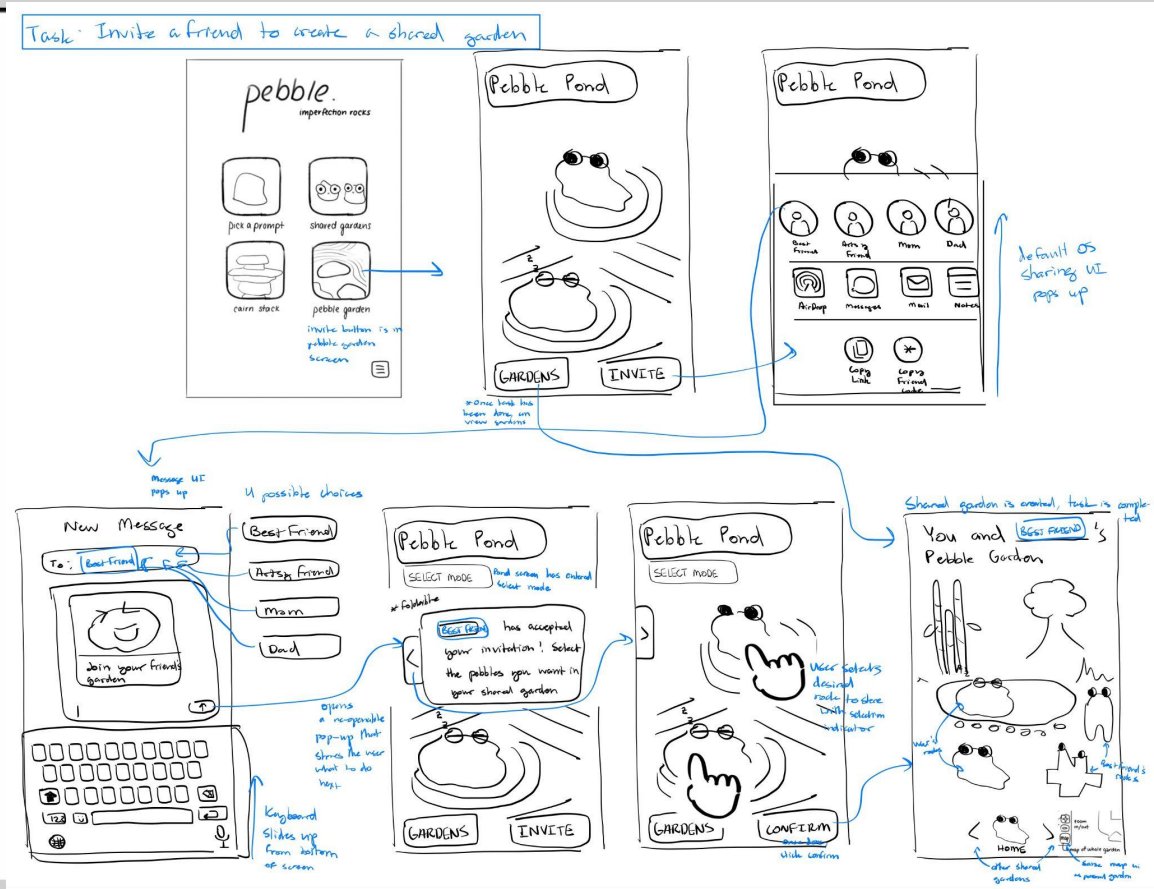
Go to shared gallery

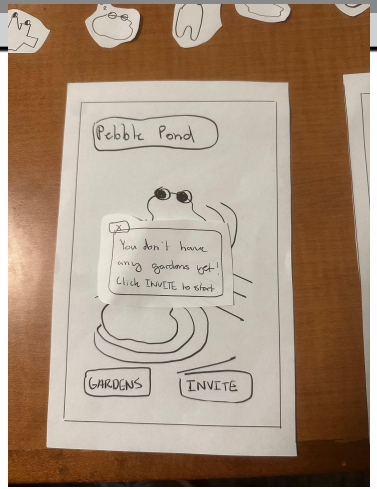
Check out a friend's drawing

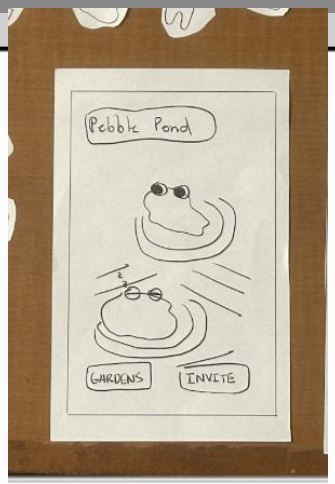


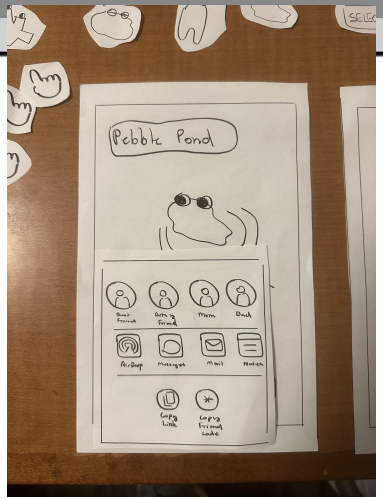
Send a message to a friend

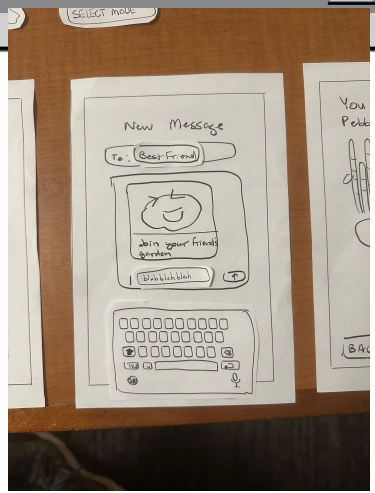
SLIDESGO

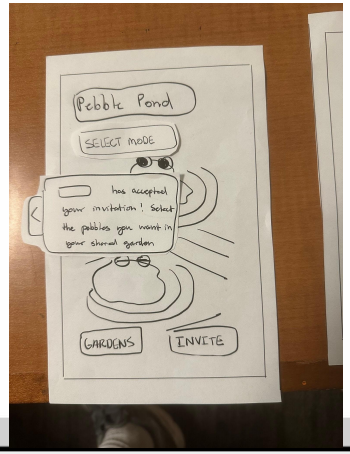

Task: Invite a friend to create a shared garden
pebble.
imperfection rocks
pick a prompt
shared gardens
cairn stack
pebble garden
invite button is in pebble gardens screen
Pebble Pond
GARDENS
INVITE
Pebble Pond
Best Friend
Artsy Friend
Mom
Dad
AirDrop
Messages
Mail
Notes
Copy Link
Copy Friend Code
default OS sharing UI pops up
*once task has been done, on other gardens
message UI pops up
4 possible choices
New Message
To: Best Friend
Join your friend's garden
Best Friend
Artsy Friend
Mom
Dad
Keyboard slides up from bottom of screen
Pebble Pond
SELECT MODE
Pond screen has entered select mode
*optional
Best Friend has accepted your invitation! Select the pebbles you want in your shared garden
opens a non-openable pop-up that shows the user what to do next
GARDENS
INVITE
Pebble Pond
SELECT MODE
GARDENS
CONFIRM
User selects desired rocks to show DARK selection indicator
while confirm
Shared garden is created, task is complete
You and BEST FRIEND 's Pebble Garden
user's rocks
best friend's rocks
HOME
other shared gardens
Journey of whole garden
Journey map in personal garden

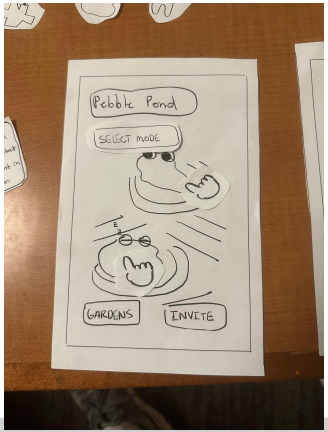


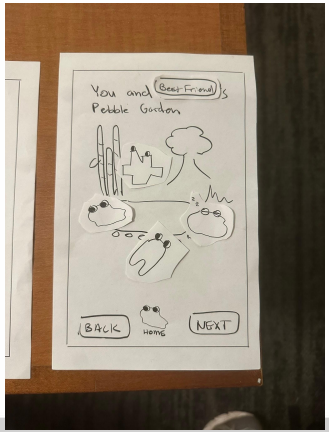










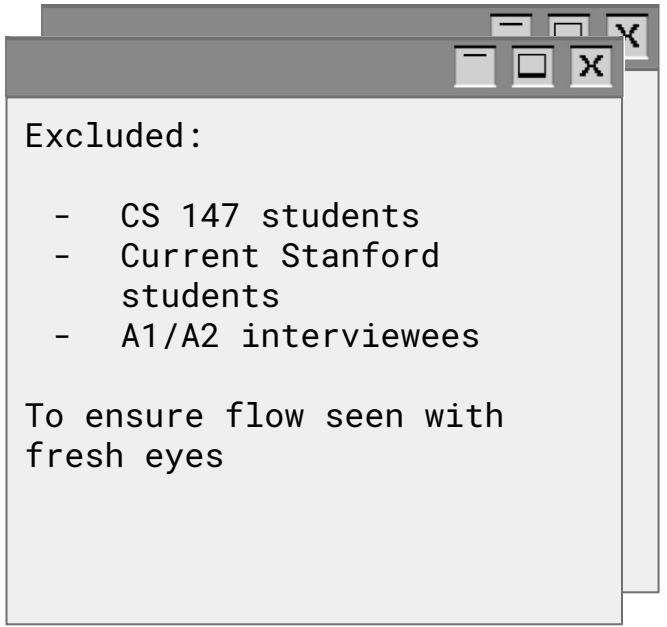


Shared Garden

Testing Methodology

Target Demographic and Participants

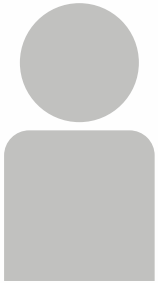
Busy students / working adults who want a creative outlet but say they lack the time



Participants

Participant 1

- Stanford alumni
- Current med student at Yale
- Digital art hobbyist



Why him?

The assumed demanding work associated with being a medical student + his hobby of drawing comics on the side.

Surveyed Tresidder during lunch and found him, and compensation was given as a free drink of choice.

Environment: sat at an outdoor table @Tresidder

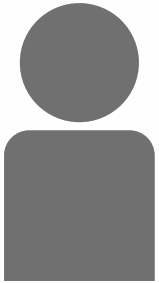
→ outdoor = quieter than inside,

→ enjoys taking breaks outside, said it aided in immersing into the mindset of ‘taking a break’.

Participants

Participant 2

- Stanford alumni
- Full time researcher at School of Medicine
- Enjoys microbiology art



Why him?

The long and unpredictable daily hours at work (sometimes staying until 10 pm at night to finish experiments) + his fun practice of creating art using microbes

Recruited around Stanford Bioengineering and School of Medicine, and offered a free meal as compensation.

Environment: table @ a café inside LKSC

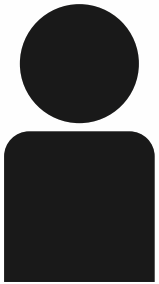
→ next to his lab, comes here often for his lunch breaks.

→ many medical students and employees also taking a break, so the ambient environment matched our testing conditions

Participants

Participant 3

- Stanford alumni
- ML engineer at a lab
- Previously acapella singer



Why him?

Since moving from student life to working life, he no longer has time for creative hobbies like drawing and a cappella.

Found through a friend. Compensated with Molly Tea from off campus.

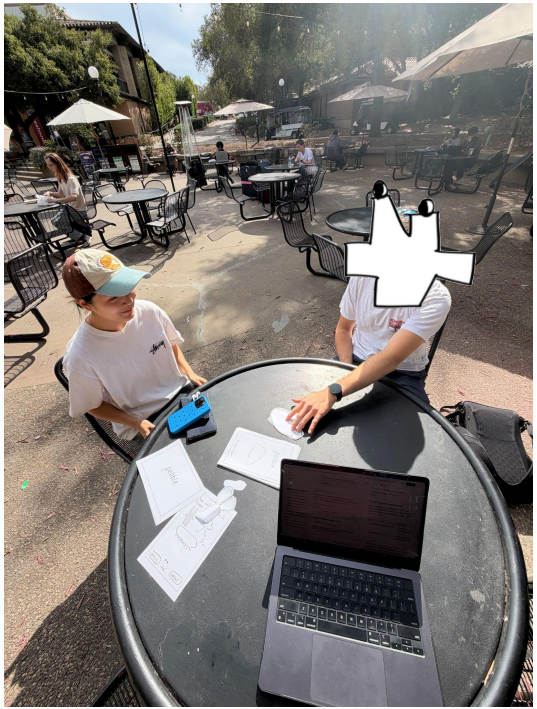
Environment:@ their partner's dorm → visit often to hang out and spend time together

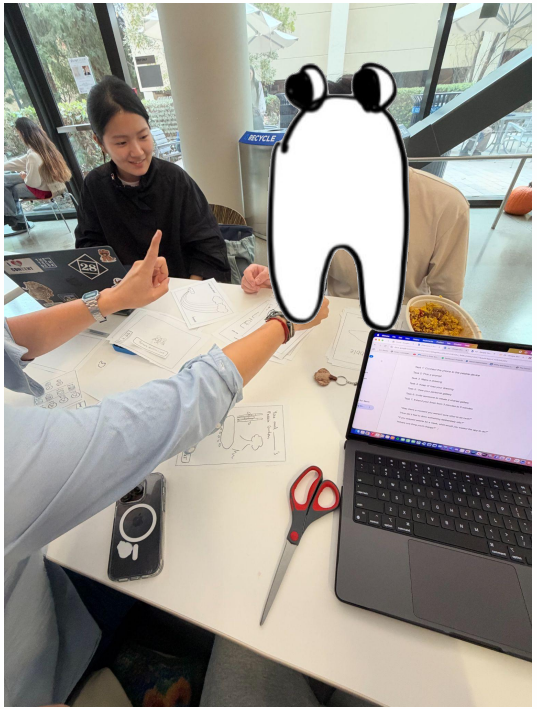
→ The dorm = relaxing and like a second home

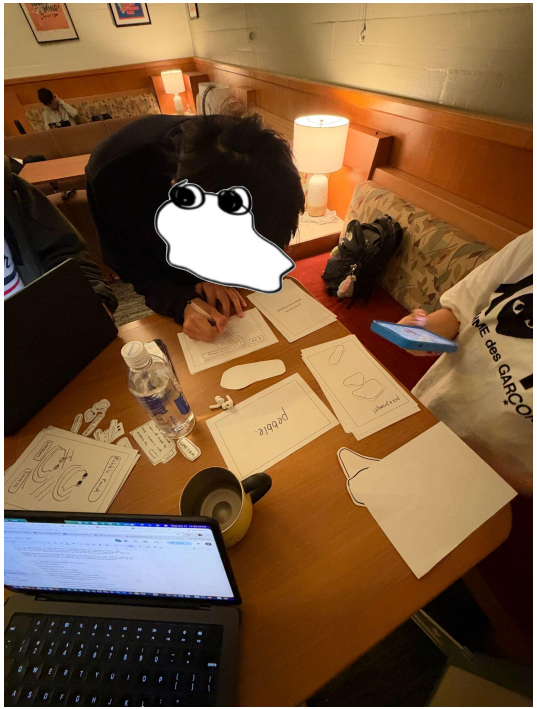
→ Tested close to 10:30 pm, so lounge was quiet

→ Lounge simulated a place a busy person would want to unwind at after a long day

Environment and Apparatus







Member Roles

Brian

Facilitator:

- Introduced the test
- Provided context and guided participants through tasks
- Concluded test with questions about the participant's experience

Ryan

Notetaker:

- Sat across from the participant and recorded observations
- Noted participant responses during post-test questions

Cherie

Computer:

- Sat beside participant and operated the prototype in response to their actions
- Did not assist with tasks

Procedures

~ 20 minutes of participant's time

Contextualize user: class, product purpose, scope of test, consent form

Made clear: testing the prototype, not the user → think aloud of what you are feeling

Communicate what to expect from us:

- Complete a task, but won't tell how
- Transition to subsequent tasks upon completion
- User can interact with the prototype naturally; computer will update appropriately

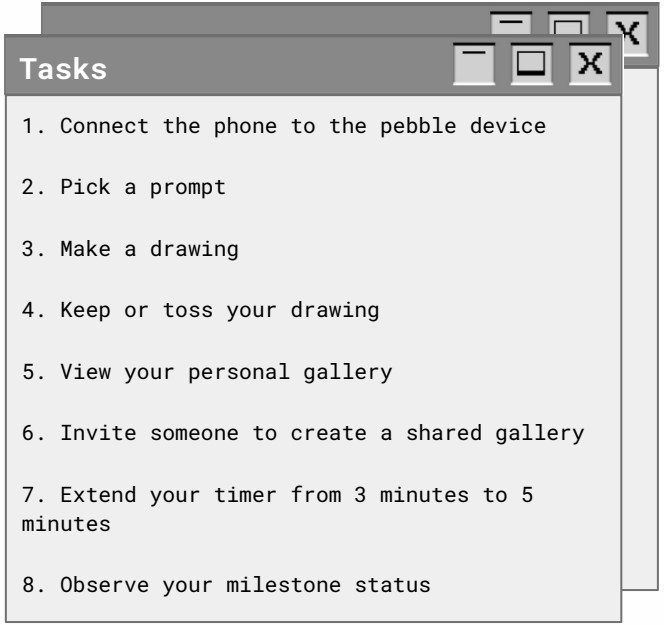
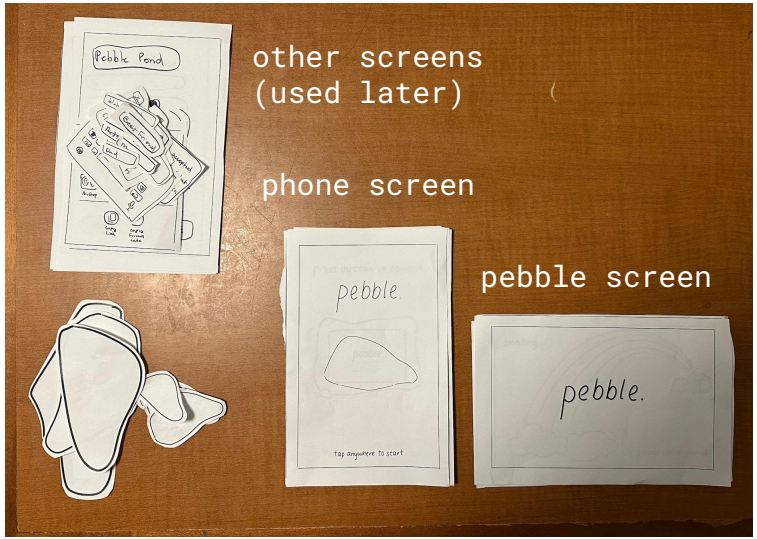


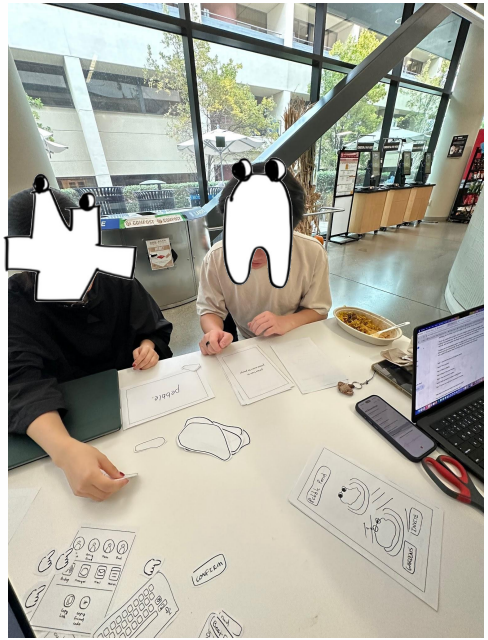
Photo of Setup

Initial Screen Setup



Test Setup

computer



participant

facilitator

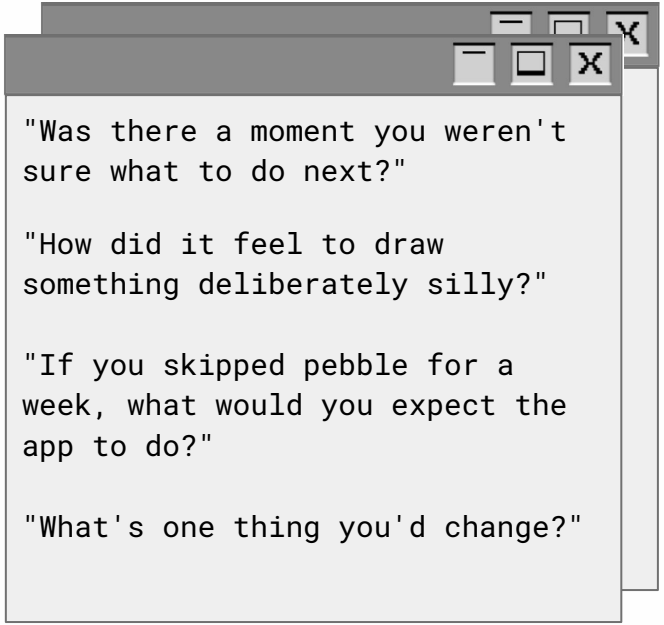
notetaker

Procedures

Upon completion:

Follow-up debrief → ask questions (shown on the right)

Ask for ratings of key measurements (explored further in usability goals slides)



Usability Goals and Key Measurements

Goal 1: Pleasing

Pebble exists to give busy people a break, so a session should leave them calmer than it found them

Key measurement: change in self-rated relaxation (1 to 5) from before key task and after key task.

Target: no rating drops

- **Process data (observational):** signs of ease vs. signs of stress; think-aloud and debrief comments on how the session felt.
- **Bottom-line data (discrete):** relaxation rating before the test and after each key task

Usability Goals and Key Measurements

Goal 2: Discoverable

Many different tasks in prototype → participants should find each feature and explore intuitively

Key measurement: errors spent per task

Target: 0 on Task 1 to 4 (the core sequence), at most 1 on each other task

- **Process data (observational):** where they look or hover first; detours to the wrong screen; labels they misread; comments such as "where would that be?"
- **Bottom-line data (discrete):** # of assists, # errors per task, and completion (unaided, assisted, or failed)
- **Bottom-line data (time-based):** time spent on task

Testing Results

What worked:

- Participants completed the initial task flows with minimal confusion
- Silly prompts lowered the pressure on drawing
- Timer encouraged participants to continue drawing, even if they finished before the timer ended.
- The prospect of sharing with friends gave participants a bit more motivation to draw

What didn't work:

- The timer was also a source of stress → a way to end early if done
- Unclear how to connect the pebble device to the phone at the start
- Most friction came from navigation
- Missing some navigation buttons (e.g., home, back)

Usability Goal 1: Pleasing

Participants saw an increase in their self-rated relaxation after the overall testing experience

Participant 2 and Participant 3 reported verbally a small spike in stress after seeing that their drawing was timed. However, their relaxation scores did not drop.

- Observed amazement and positive remarks at the cairn stacking milestone tracker
- Many giggles and laughters during drawing
- The rock character drawings also led to increased laughter
- All participants decided to keep their drawings

Usability Goal 2: Discoverability

Most tasks were completed correctly.

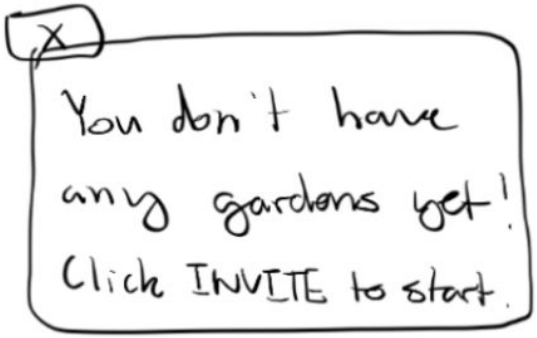
Participant 2 and participant 3 both made an error at Task 1, which was connecting the phone to the pebble device.

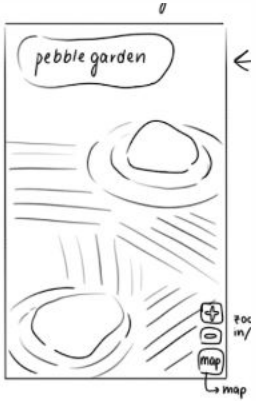
Participant 3 struggled with finding out how to invite a friend to create a shared pebble garden.

- Even after the ‘connecting device’ screen instructed users to touch the device, participants regardless overlooked
- Participant 3 clicked repeatedly on the same button to comprehend how to get to shared pebble garden, to no avail
  - Led to longest time spent on a task, stuck in the home screen for ~30 seconds

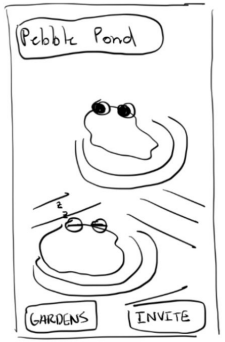
#1 Critical Incident

UI Mismatch with Inviting Friends to Shared Garden





Intended Button ->
<- No Invite Button



Prompted user to click “invite” on rock garden screen with no such “invite” button

Level 4 of confusion for user

Discussion

Implications:

Implication 1
Users shouldn’t need to guess anything → more visible, intuitive UI that doesn’t disrupt user flow

Implication 2
Having too much time can also add stress → give users the choice/freedom to end earlier

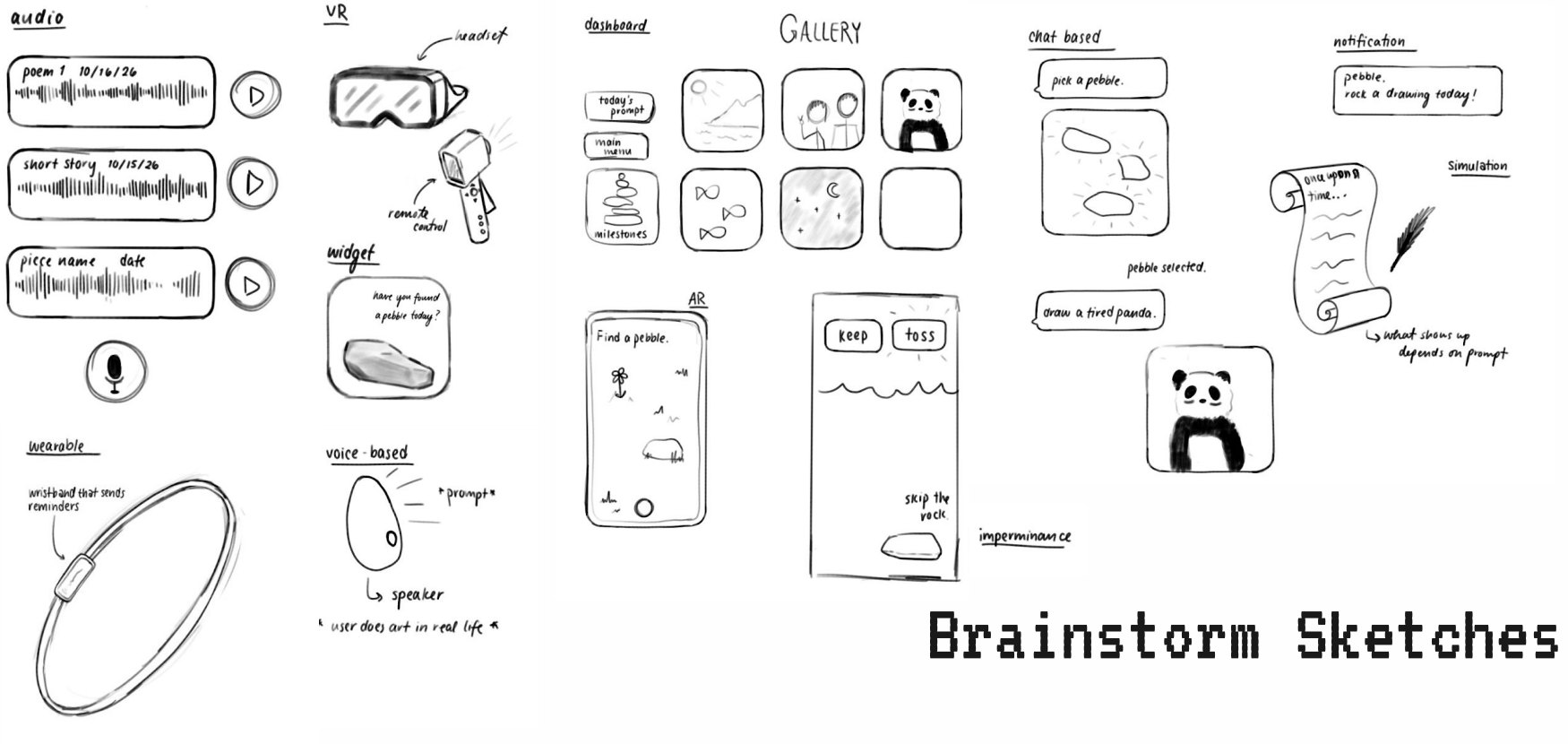
Implication 3
Simply having a second interface can create confusion despite efforts to make instructions clear

Implication 4
Even if drawings aren’t perfect, people find satisfaction in keeping record of what they’ve drawn

What Will We Change?

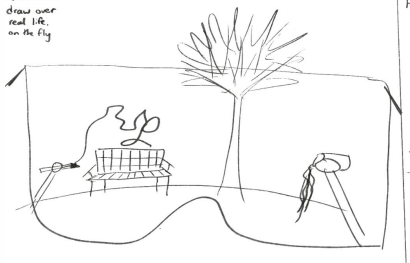
- During connection to pebble device, add explicit instructions to look at device screen
- Add consistent navigation buttons to prevent guessing
- Pivot access point of inviting friends into shared garden screen, where it appears more intuitive
- Add an end-early button during drawing
- Add small, interactive features (pop the bubbles, make the fountain flow in the garden) based on participant 1’s feedback

Appendix

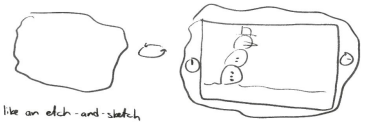


Brainstorm Sketches

AR Glasses
draw over real life on the fly



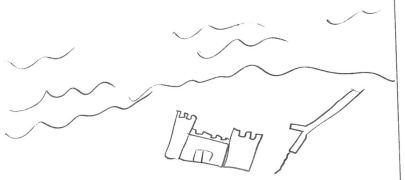
Pebble - companion



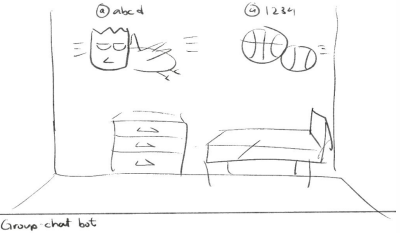
- like an etch-and-sketch
- can be linked to your phone
- carry the pebble, vibrates in pocket
- displays prompt, whatever is drawn is conveyed to app

VR - beach: use a stick to draw on the sand

- wave acts as a timer
- at time, fossil is made (?) it

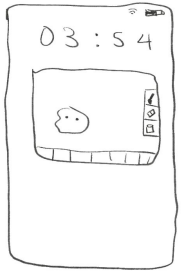


AR/VR shared room
- upload your drawing to server, can visualize your friend's drawings float in your room

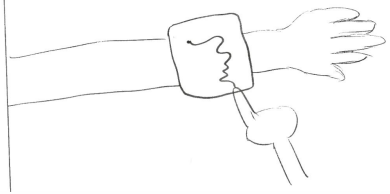


Lock screen pebble

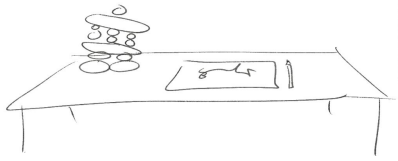
- draw straight on lock screen widget
- no app to open



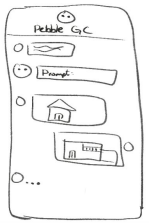
- finger doodle on a small canvas (watch) so that imperfection is guaranteed



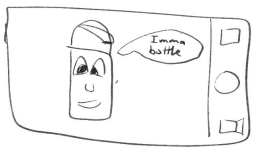
AR rock garden
- pebbles appear as 3D stones stacking on your desk
- milestone as visible heights



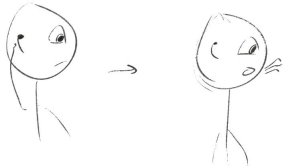
Group chat bot
- drops a prompt into an existing friend chat; people can choose to reveal



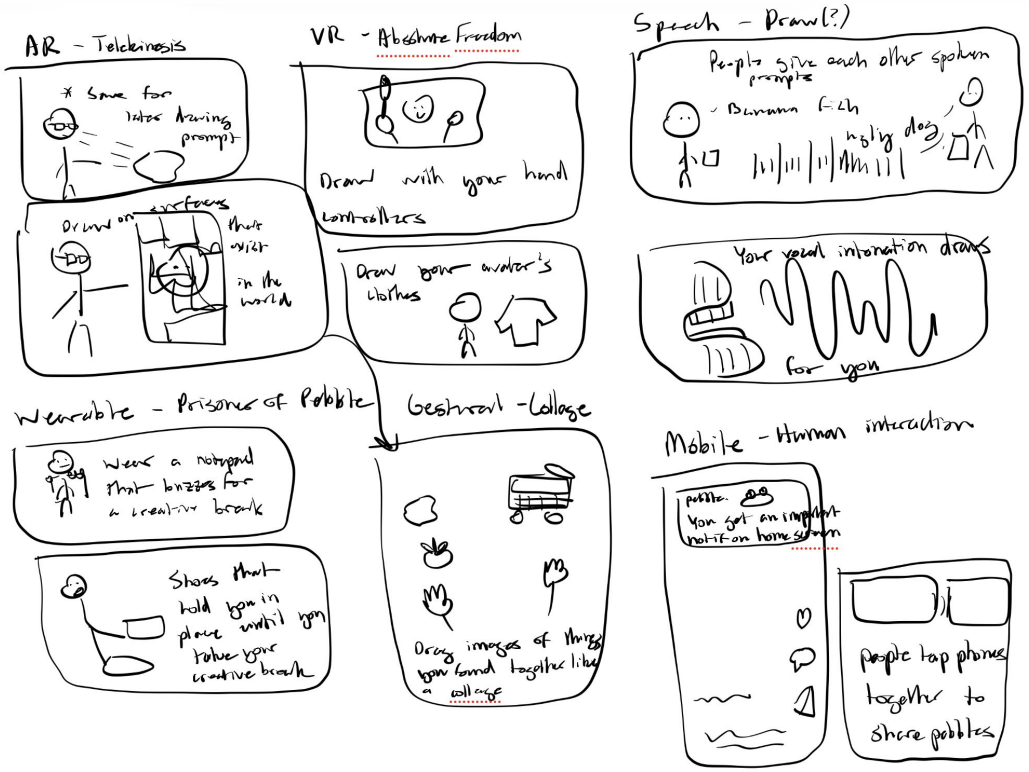
Draw faces - AR
- draw googly eyes & speech bubbles on real objects through camera
- photo = pebble



Earbud Pebble: double-tap earbud to hear a prompt
→ answer out loud while walking
- one take, no redos
(more poem focused)

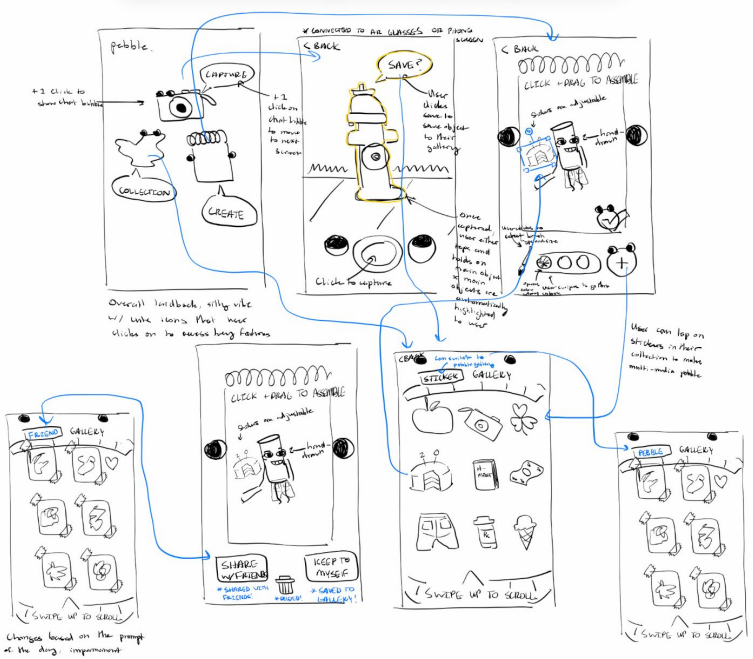


Brainstorm Sketches



Brainstorm Sketches

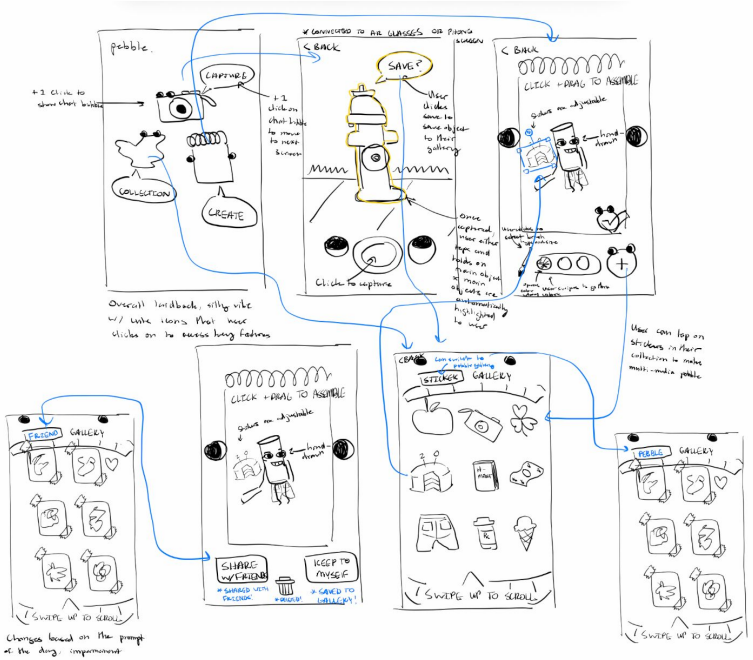
AR Sticker Collage



Pros

- interactive with the world (taking photos)
  - Augmented reality aspect allows for user intentionality with real world inspiration
- reusable icons/motifs – people can go back and reuse stickers they created previously
  - Stickers can be anything the user wants
- Allows user to share their drawings/collages with friends/engage in social interaction
- Gallery option allows users to look back on past collages/drawings; power/sense of accomplishment in milestones/scrolling through several collages
- Low barrier to entry: users can just use their camera app to take photos and make a collage

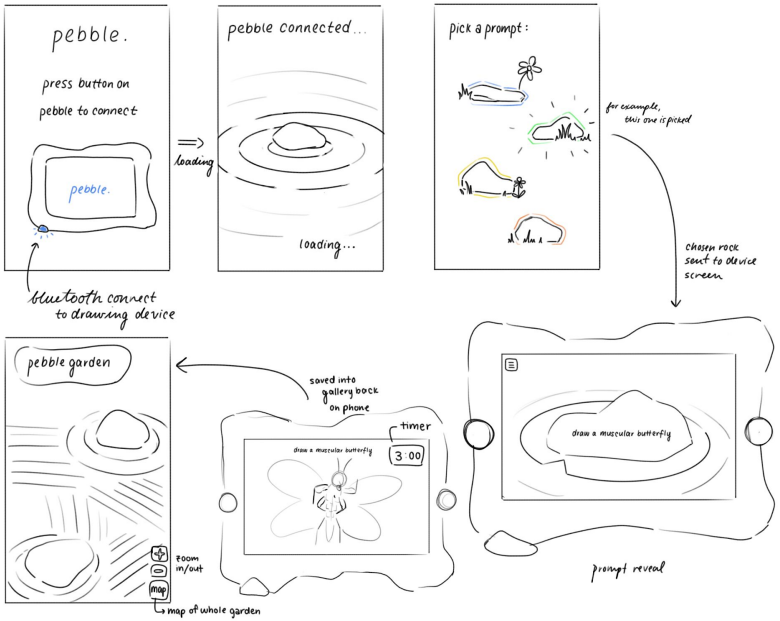
AR Sticker Collage



Cons

- harder to do actual drawing on the phone/maybe users will feel more incentivized to only do the photos -> collage which limits modes of creativity
- everything seems very visual, maybe accessibility concerns
- Stickers can be anything the user wants, so how do we make sure that there are no bad actors/bad actors get nullified before they can act badly
- (niche) User might have awkward interactions where they are inspired by a person's bag and have to ask them to take that picture
- possibility of confusion in the intuitiveness of the design if interface seems too "busy"/"crowded"
- Interface is not revolutionary; essentially a dedicated drawing-sharing app
- Doesn't fit nicely into needfinding or our tasks or our concept video

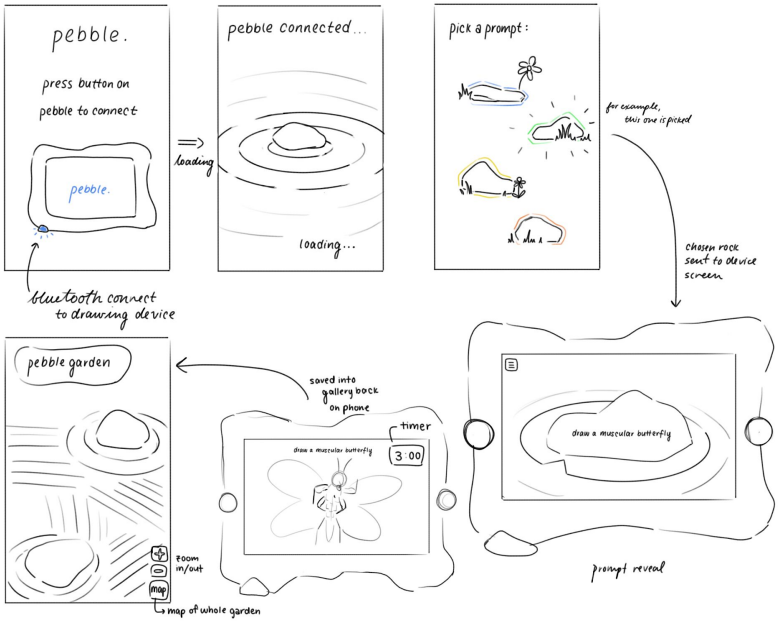
Physical Device



Pros

- Fits needfinding quite well/better than other realization (more of a refuge)
- Physical device stops user from getting distracted by phone while engaging in refuge
- Specifically uses timer and randomizes prompts which gives user lower stakes for what they draw
- There is no mentally ‘demanding’ tasks based on the simplicity of the design → all the user has to do is pick and draw, so there is less pressure to make anything demanding

Physical Device



Cons

- seems like it is limited to drawings only / unnatural to add other forms of art (ex. poetry…?)
- No social aspect, but definitely addable
- requires a whole other device as opposed to everything being central to one device
- Doesn't necessarily NEED a whole other device other than the goal of intentional mindfulness and a refuge from other responsibilities that might be present on a phone

Why pebble device over AR? (expanded in speaker notes)

AR app: its cons hit the core loop

Drawing on a phone is awkward → drift photo-to-collage (assembling instead of making)

Hunting for real-world inspiration → effort and social awkwardness (opposite of low-stakes)

Without the AR layer → standard drawing-sharing app that maps poorly to our needfinding and tasks

device: its cons are peripheral

Missing social layer was addable → shared pebble garden and invite flow

Drawing-only scope fits the quick, deliberately silly sketch premise

Randomized prompts and a short timer lower the stakes

trade off we accepted

An extra device isn't technically required

Physical separation *is* the refuge → users off the device where their work lives

A second device for a protected break ✓

AR: richer capture and social features at the cost of the break

Note on Technology Apparatus

A phone to record the interaction + a computer to take notes and read the script from.

Other than for these purposes, **no technologies** were used; all the prototypes were printed in **paper**. We used a double sided tape to incorporate interactivity with the user and to add dynamic features.

Testing Script

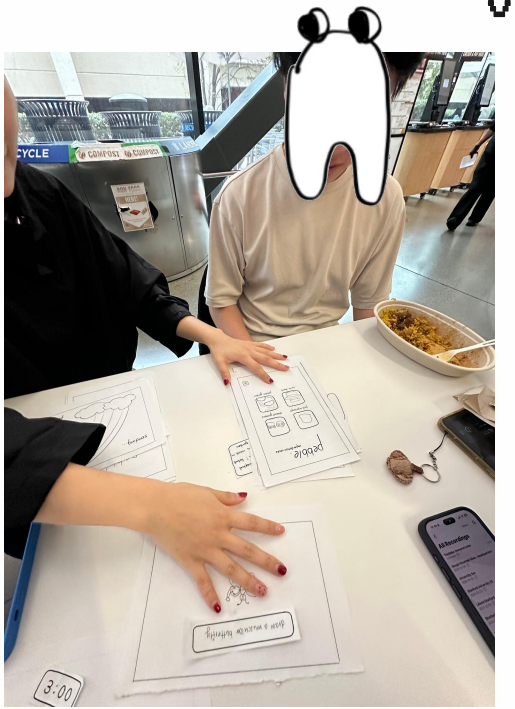
"Hi, thanks for helping us. We're Stanford students designing a phone app and complementary device called pebble, and today you'll try an early paper version of it. We're testing the app, not you. If anything is confusing, that's our fault, and it's exactly what we want to find."

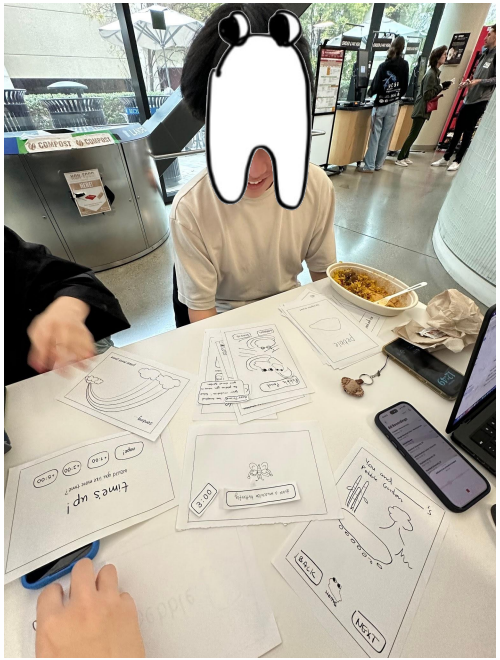
[Hand over consent form.] "[**Notetaker**] will take notes and a few photos of your hands and the prototype, and no photos of your face unless you say it's okay. You can stop at any time."

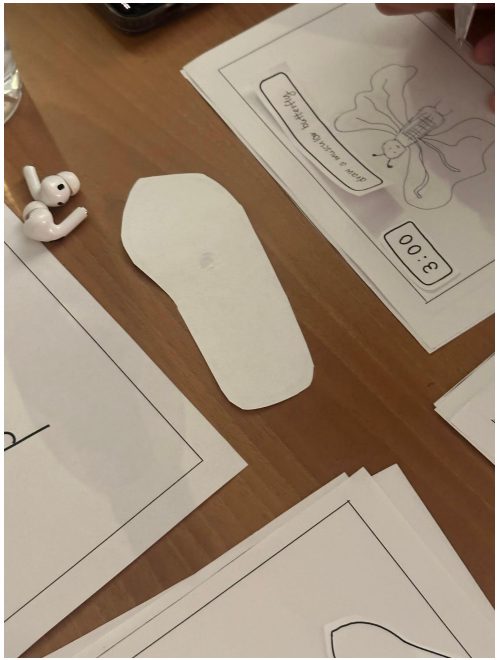
"As you go, please think out loud: what you're looking at, what you expect to happen, and anything that surprises you. Tap the screen like a real phone and we'll change what you see. [**Computer**] can't talk, and I can't explain how things work, but I may ask what you're thinking."

"Imagine you just downloaded pebble. Your days are full of classes or work, and you got the app because you miss making things. We will ask you to complete a few tasks, but we won’t tell you how; a part of this testing relies on whether users can complete tasks intuitively."

Other Photos From Testing







Key Usability Measurements: Pleasing

"How relaxed do you feel currently?"

(5 is most relaxed, 1 is least relaxed)

Participant 1

Relaxation Rating

Before test: 3

Connecting devices: 4

After drawing: 5

Sharing with friends: 5

After test: 5

Participant 2

Relaxation Rating

Before test: 2

Connecting devices: 3

After drawing: 2

Sharing with friends: 3

After test: 4

Participant 3

Relaxation Rating

Before test: 3

Connecting devices: 4

After drawing: 4

Sharing with friends: 4

After test: 4

Key Usability Measurements: Discoverability

"How many errors on each task?"

Participant 1

1. 0
2. 0
3. 0
4. 0
5. 1
6. 1
7. 0
8. 1

Participant 2

1. 1
2. 0
3. 0
4. 0
5. 0
6. 2
7. 0
8. 0

Participant 3

1. 1
2. 0
3. 0
4. 0
5. 1
6. 5
7. 2
8. 0

Critical Incidents Log

Participant 1

Task	Critical Incident	Rating	Notes
Connect phone to pebble device	Participant initially pressed the pebble button on the phone screen instead of the physical button on the device.	**2**	Seemed unclear at first, but quickly realized after seeing that the computer didn't change anything.
Drawing	Continued drawing until the timer was done	**2**	Can be easily addressed with a small design change (i.e., "finished early" button)

Critical Incidents Log

Participant 2

Task	Critical Incident	Rating	Notes
Connect phone to pebble device	Didn't know the connection required pressing the physical device	**3**	
Drawing	** NOT AN INCIDENT ** but no way to indicate finishing drawing early other than verbal	**2**	Can be easily addressed with a small design change.
All	Said what he wanted to happen rather than clicking to make it happen	**2**	Likely because a person was operating the "computer." Will be resolved in digital prototype.
Picking Prompt	"Do I flip this?" when he selected a rock	**2**	The paper prototype's flippable elements invited a different interaction style than a digital screen would. A limitation of the paper prototype, not necessarily a usability issue with the design.
Looking at Personal Drawing	No back button on the drawing screen made it hard to navigate back after finishing.	**3**	

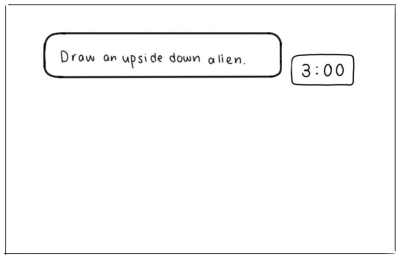
Critical Incidents Log

Participant 3

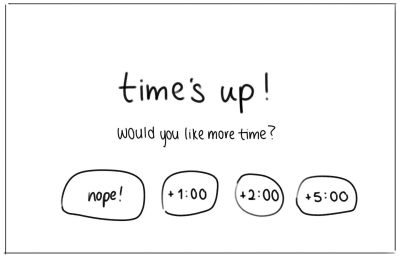
Task	Critical Incident	Rating	Notes
Connect Device and Complete a Drawing	Didn't click the button on the pebble to connect but did adequately click on the pebble's "screen."	**3**	
Complete a Drawing	Kept looking at timer and decided to draw more when he had a lot of time even when he perceived his drawing as done	**2**	
Change the Timer Setting	Intuitive but noted that the settings tab had really small font	**1**	Small text may hurt readability but this can be addressed with an easy design change.
Change the Timer Setting	Saw settings button as the same as the home button, hard to navigate the UI	**2**	
Create and Explore a Shared Garden	Didn't know where to find the invite button; "Click [Invite]" on the UI but no Invite button on the screen	**4**	

Task Flow: Complete a drawing

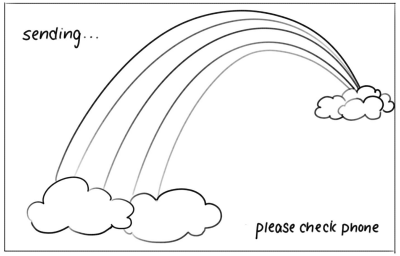
Draw on pebble device



Choose to extend timer

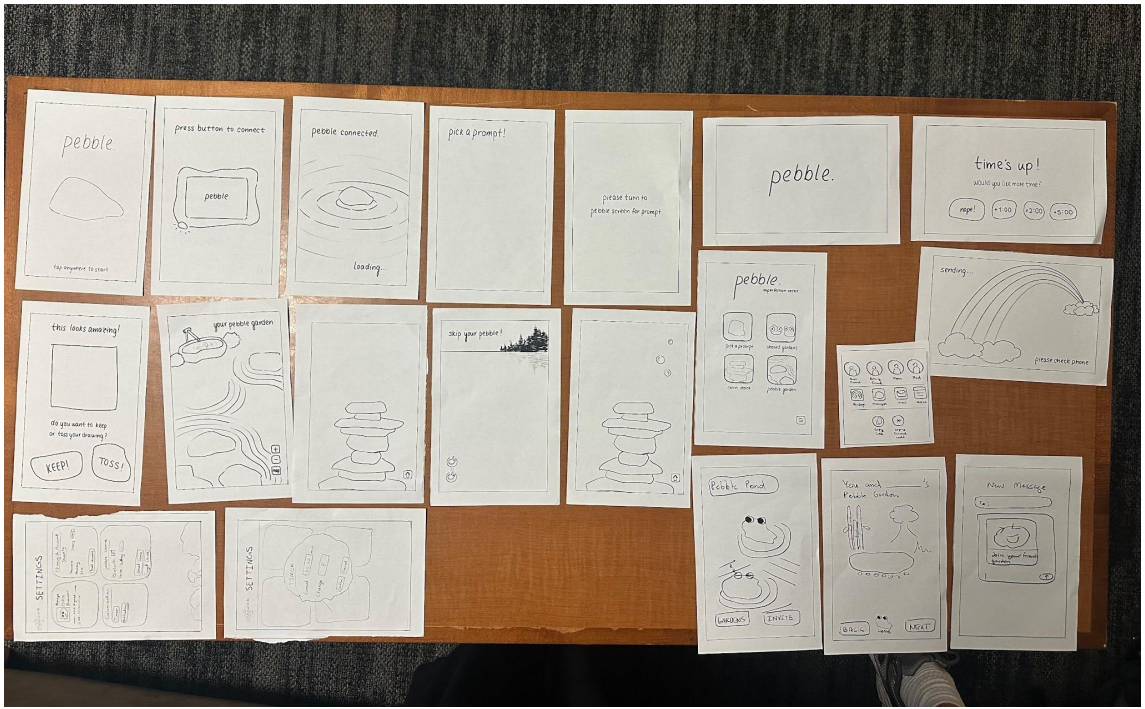


Finish drawing

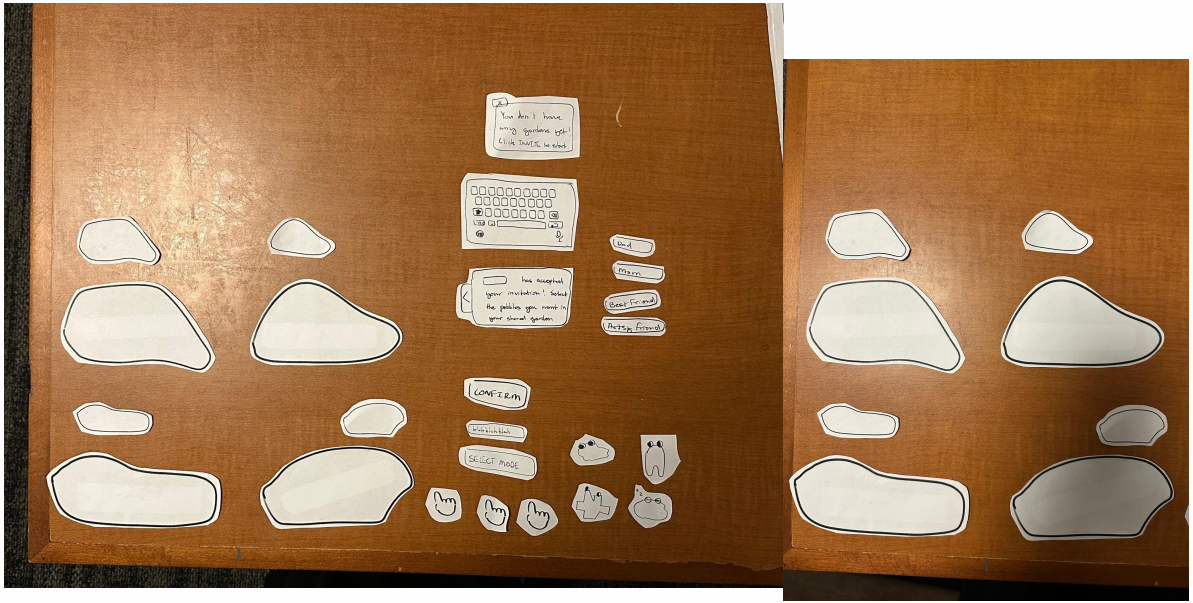


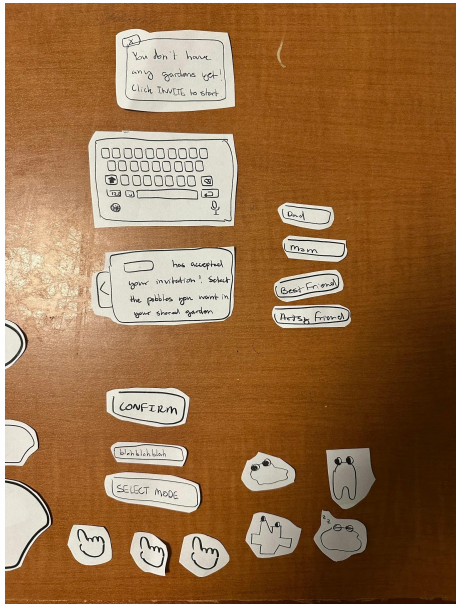
Paper Prototype Additional Photos

Bird's eye view, all screens

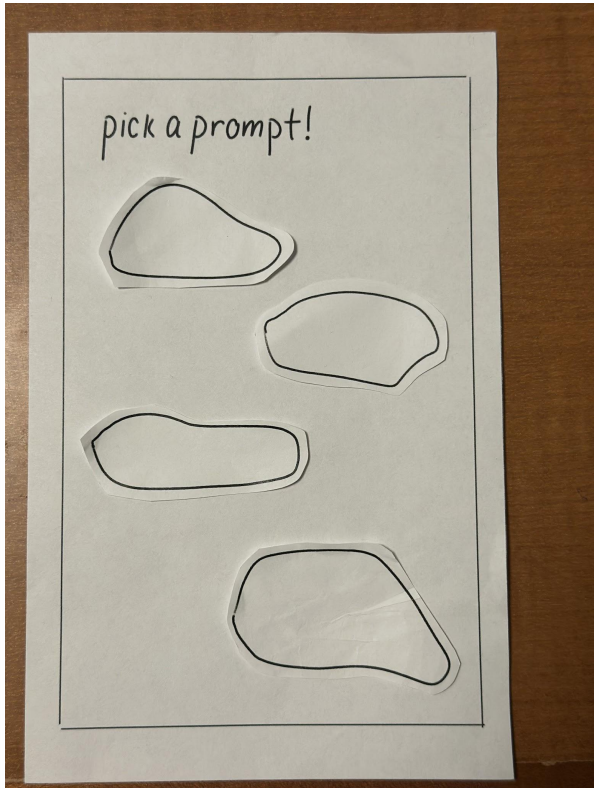


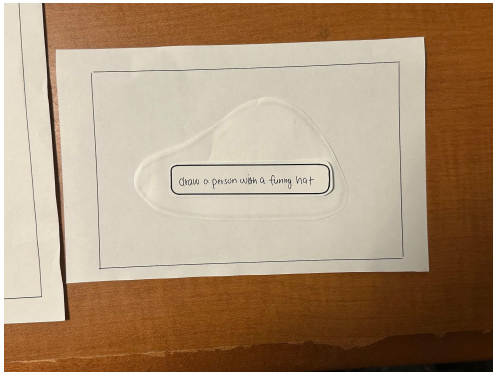
Bird's eye view, Interactable Objects

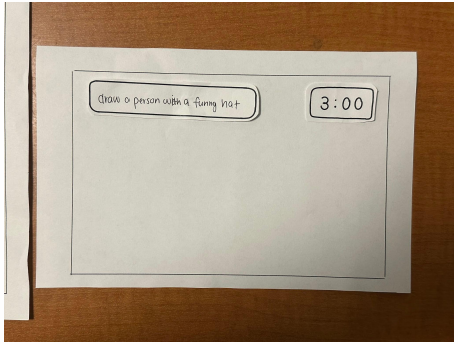


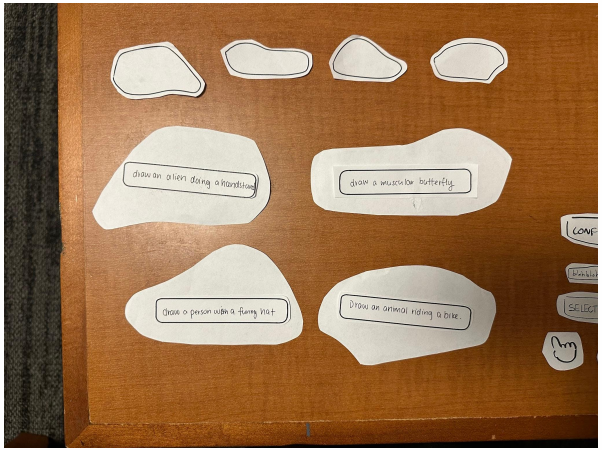


Picking a prompt to draw

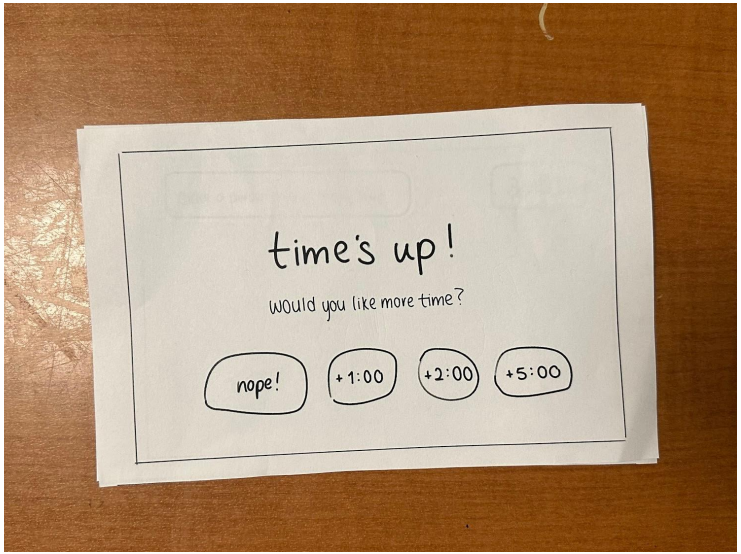


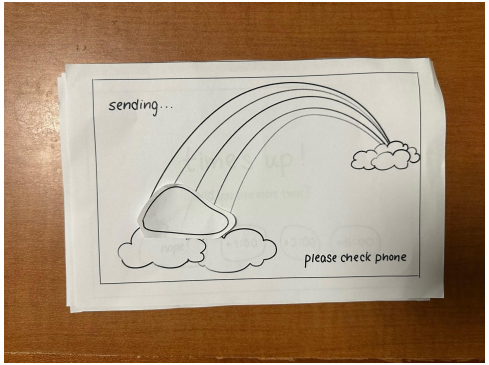


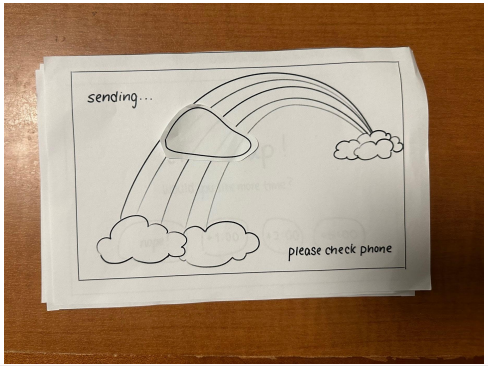




Adding time or sending drawing to phone from Pebble

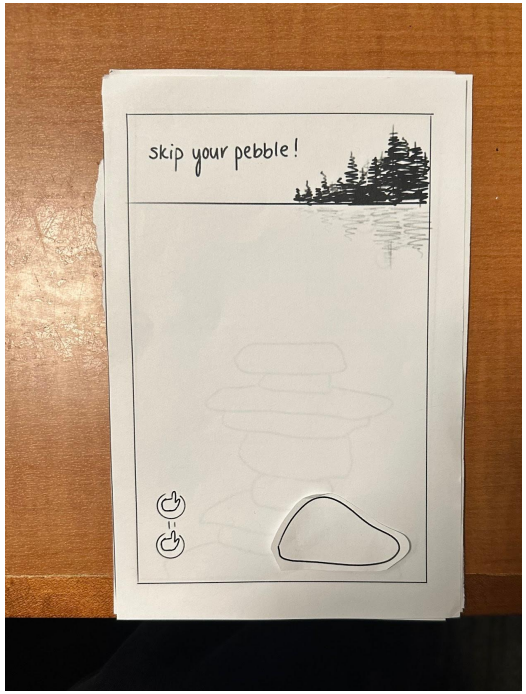




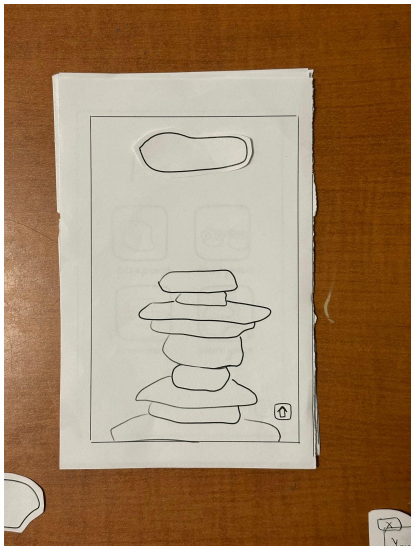


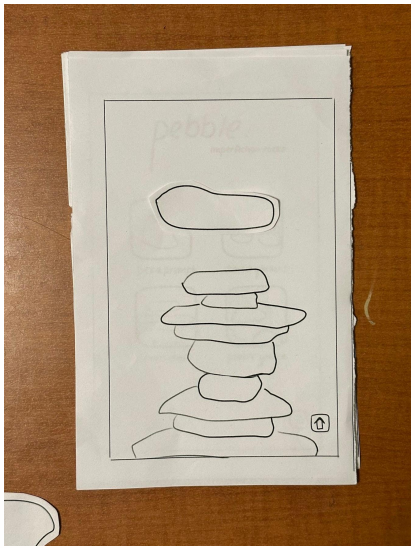
Adding rock to garden

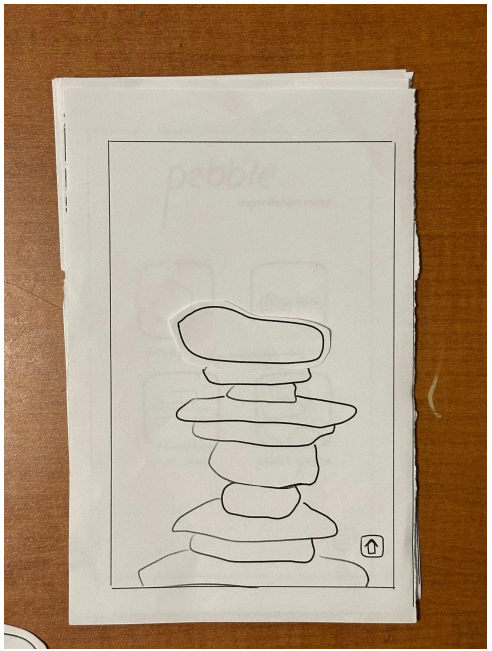
Skipping pebble when tossing



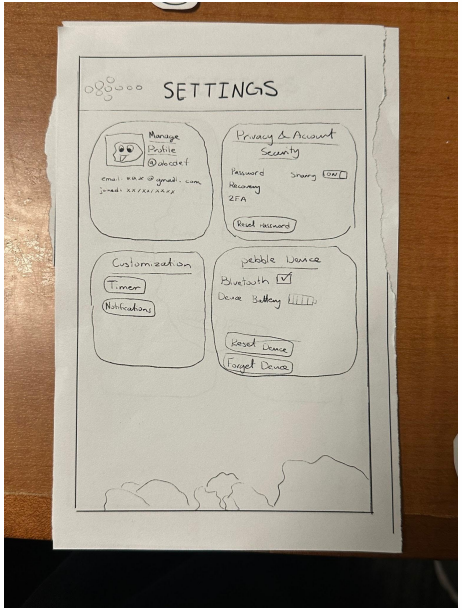
Cairn Stack (Pebble sinking when tossed)

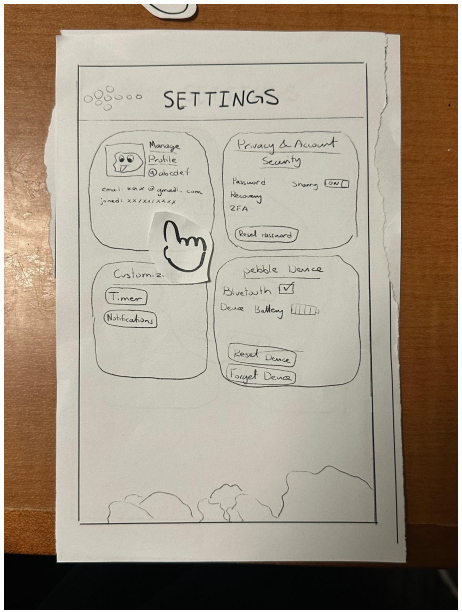


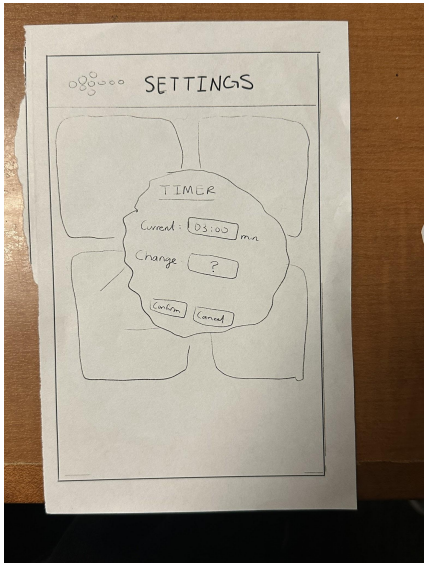




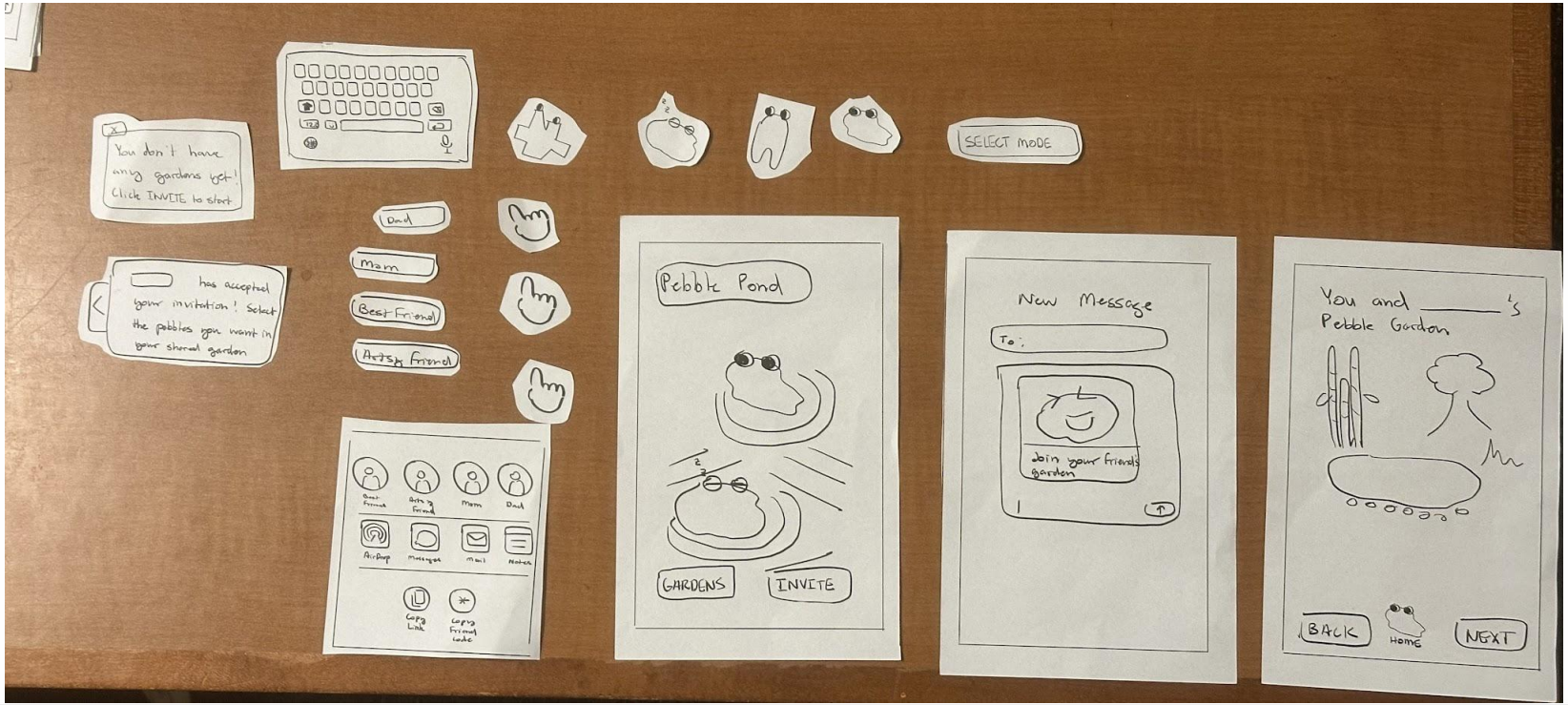
Changing the timer



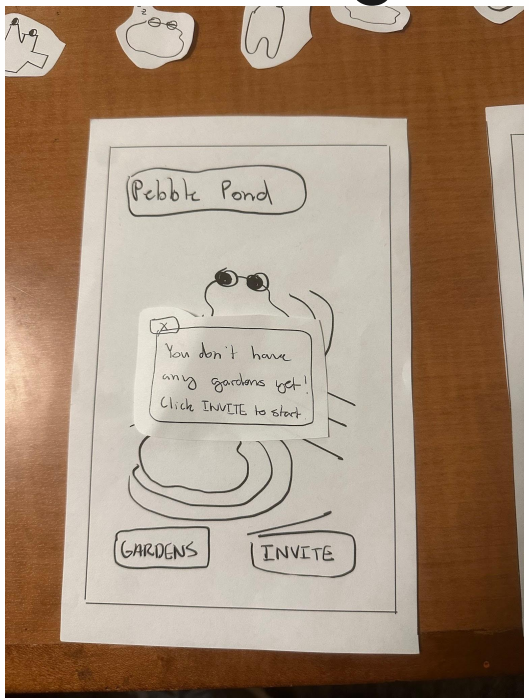


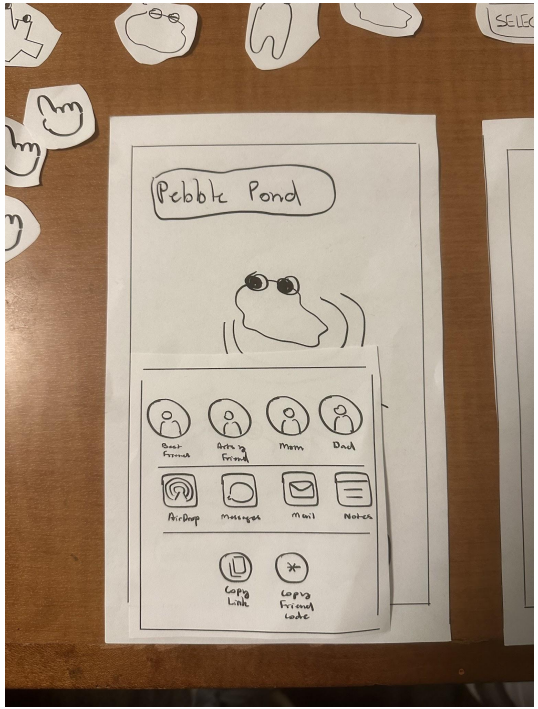


Inviting Friend to Shared Garden

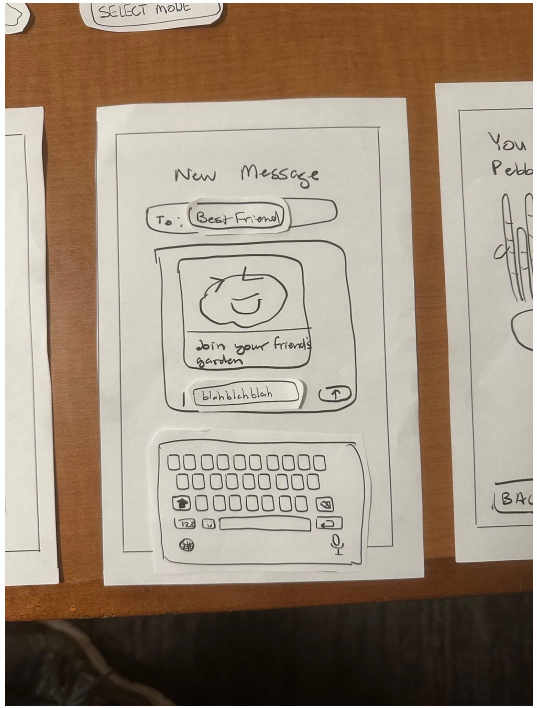


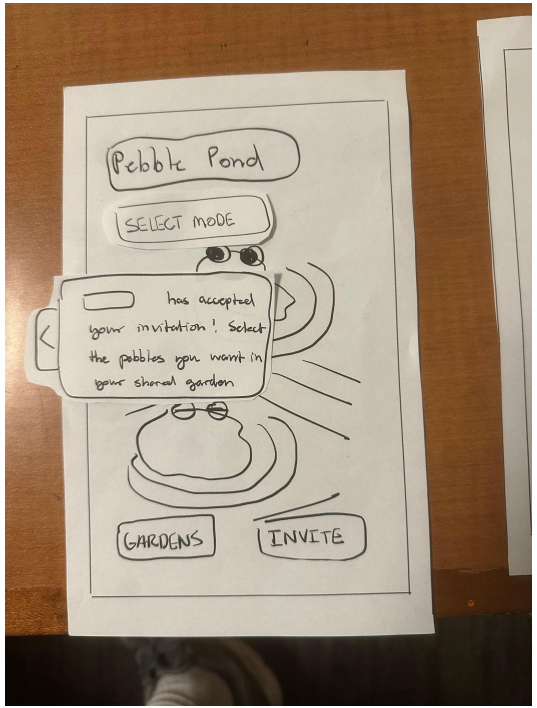
Inviting Friend to Shared Garden

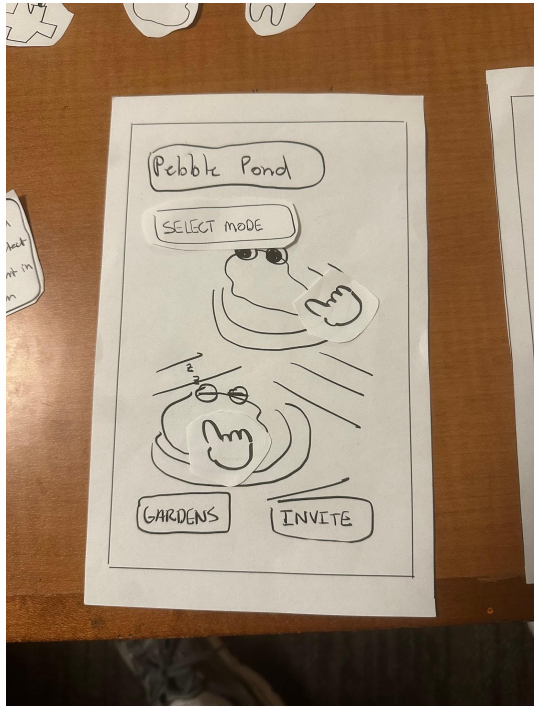




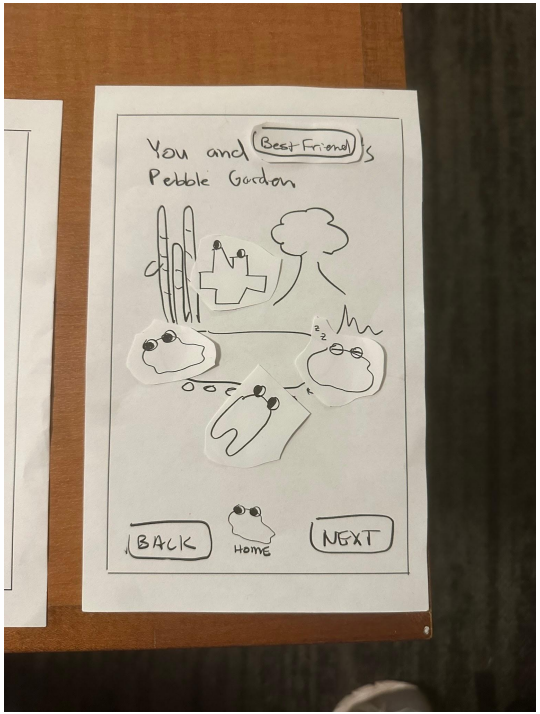
Inviting Friend to Shared Garden

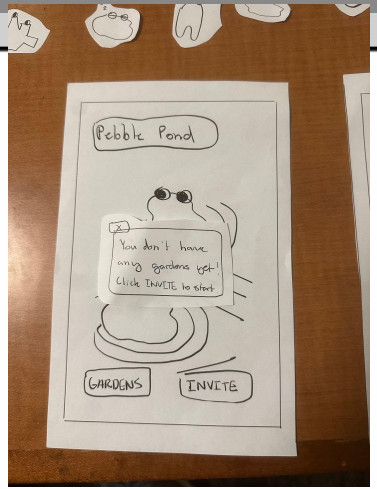


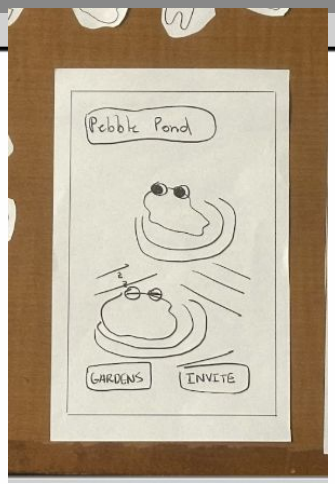


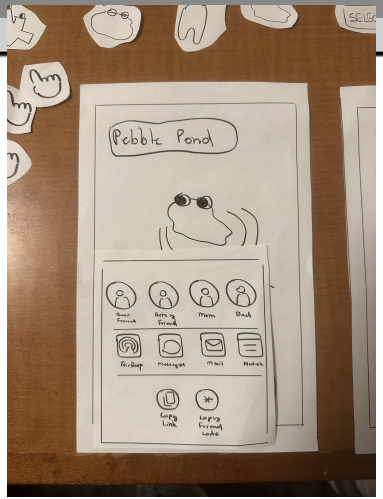


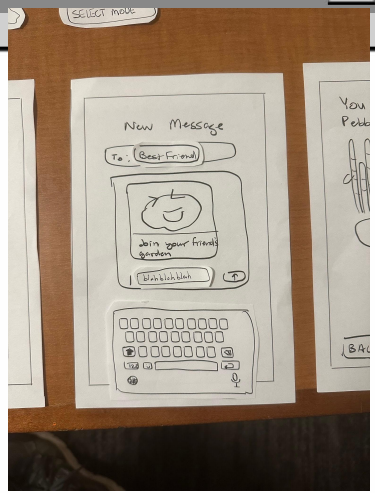
Inviting Friend to Shared Garden

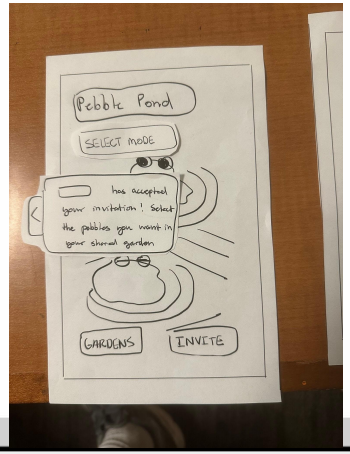


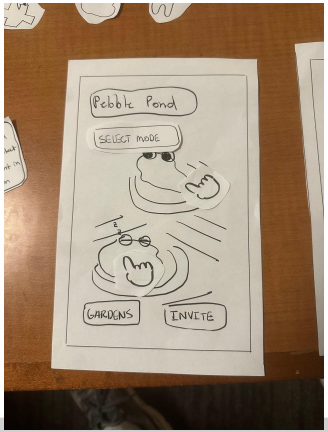


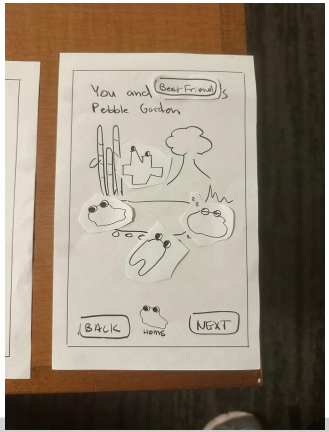












Full Flow, Shared Garden